

PARTS MANUAL

**S1.6 - BR
(A259)**

CUSTOMER EDITION

HYSTER

SECTIONS

1	FRAME/DRIVE UNIT
2	ELECTRICAL SYSTEM
3	LOAD WHEEL
4	HYDRAULIC SYSTEM
5	MAST-TWO STAGE LIMITED FREE-LIFT
6	MAST-THREE STAGE FULL FREE-LIFT
7	CARRIAGE
8	NUMERICAL INDEX

This Parts Manual is divided into 8 major sections which are listed above. Quick reference to these sections can be made by placing the right thumb on the tab of the desired section, bending the book back and thumbing the pages to the corresponding tab.

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ABBREVIATIONS QUICK REFERENCE SHEET

Listed below are abbreviations used in parts manuals to denote the following expressions:

TERM	DEFINITION	TERM	DEFINITION
Amp	Amperes	FRE	Freezer Environment
AR	As Required	in.	inches
BDI	Battery Discharge Indicators	LPG	Liquefied Petroleum Gas
BWI	Brush Wear Indicator	MAP	Manifold Absolute Pressure
CE	Communicator End	mm	millimeters
CRE	Corrosive Environment	MSTS	Microprocessor Spark Timing System
D/B	Date Break	NPT	National Pipe Threads
dbA	Decibel "A" Scale	O/S	Oversize
DE	Drive End	PTO	Power Take Off
ECH	Empty Container Handler	RPM	Revolutions per Minute
ECH-3	Empty Container Handler - 3 High Stacking	S/N	Serial Number
ECH-4	Empty Container Handler - 4 High Stacking	TMM	Truck Management Module
ECH-5	Empty Container Handler - 5 High Stacking	UL Type D	Diesel-Powered Safety Rating
ECH-6	Empty Container Handler - 6 High Stacking	UL Type DS	Diesel-Powered Safety Rating
ECH-7	Empty Container Handler - 7 High Stacking	UL Type E	Battery-Powered Safety Rating
ECM	Electronic Control Module	UL Type EE	Battery-Powered Safety Rating
ECN	Engineering Change Notice	UL Type ES	Battery-Powered Safety Rating
EEPROM	Electrically Erasable Programmable Read Only Memory	UL Type G	Gasoline-Powered Safety Rating
EPA	US Environmental Protection Agency	UL Type GS	Gasoline-Powered Safety Rating
EPROM	Electrically Programmable Read Only Memory	UL Type LP	Gas-Powered Safety Rating
		UL Type LPS	LP Gas-Powered Safety Rating
		U/S	Undersize
		UL	Underwriters Laboratory
		V	Volts

HOW TO USE THE ILLUSTRATED PARTS MANUAL

This parts manual describes and illustrates assemblies, subassemblies, and detail parts needed for service replacement.

The different constructions are indicated by keys and footnotes. The callouts correspond to descriptions found on the next page.

HYSTER UNIT CODE D001 NORTH AMERICA 06/2001

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AIR INTAKE AND FILTER GASOLINE AND LPG	
ALTERNATOR GASOLINE, LPG, AND DIESEL	
ALTERNATOR TYPES GS, LPS, AND DS	
AUXILIARY FUNCTION TUBE TWO AND THREE STAGE FULL P	

SECTION 13 NUMERICAL INDEX

PART NO.	PAGE	ITEM NO.	PART NO.	PAGE	ITEM NO.	PART NO.	PAGE	ITEM NO.	PART NO.	PAGE	ITEM NO.
142	3-4	19	10-13	25	11-27	29	12-31	12	16719	3-36	5
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			11-11	6	8-4	29	16863	3-36	16863	3-36	20
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- 9 MAST-TWO STAGE
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FRAME

ELECTRICAL SYSTEM

TRANSMISSION HOUSING
⇒ S/N D001H-2628

KEY:
A: Diesel
B: Gasoline and LPG

ITEM	PART NO.	DESCRIPTION	QTY	A	B
1	2022547	Transmission	1	1	1
2	2023329	Transmission	2	2	2
3	2021670	Case	4	4	4
4	2023418	Bearing Cap	1	1	1
5	296426	Washer	2	2	2
6	293991	Bolt	2	2	2
7	2021390	Cover	4	4	4
8	1371656	Cotter Pin	1	1	1
9	2021329	O-Ring	4	4	4
10	2021915	Plug	22	22	22
11	2021675	Housing	39	39	39
12	2021336	Pin	1	1	1
13	287860	Bolt	1	1	1
14	292682	Lockwasher	1	1	1
15		Hydraulic Pump	1	1	1
16		⇒ Figure 5-3	8	8	8
17	2021345	Gasket	1	1	1
18	214364	Bolt	1	1	1
19	296418	Washer	1	1	1
20	2021388	Temperature Switch	1	1	1
21	2021389	Breather	9	9	9
22	2021383	Dipstick	2	2	2
23	292515	Bolt	3	3	3
24	136692	Nut	3	3	3
25	292527	Capscrew	4	4	4
26	292531	Lockwasher	4	4	4
27	292683	Bolt	1	1	1
28	293637	Washer	1	1	1
29	296420	Transmission Kit (Includes Item 7)	1	1	1
30	996403	Transmission Kit (Includes Item 14)	1	1	1
31	996405	Transmission Kit (Includes Item 14)	1	1	1

This Parts Manual is divided into 11 major sections which are listed above. Quick reference can be made by placing the right thumb on the tab of the desired section, bending the pages to the corresponding tab.

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HOW TO FIND THE DESIRED PART NUMBER

WHEN THE PART NUMBER AND THE NEXT HIGHER ASSEMBLY IS NOT KNOWN:

1. Determine the function and application of the part required. Turn to the Sections Page. Choose the general area of reference most likely to include the part.
2. Turn to the section you chose. Use the Section Table of Contents to determine the assembly which would normally contain the part required. Then locate the part on the assembly breakdown page.

WHEN THE PART NUMBER IS NOT KNOWN AND THE NEXT HIGHER ASSEMBLY IS KNOWN:

3. Determine the assembly the required part is used on. Turn to the Table of Contents (Page i).
4. Locate the assembly the required part is used on and turn to the page indicated for that assembly. Then locate the part on the assembly breakdown page.

WHEN THE PART NUMBER IS KNOWN:

5. Use the Numerical Index (Page 8-1) to find the part number. Turn to the page listed and locate the part as indicated by the item number.

GENERAL:

The assembly breakdowns include part numbers, descriptions, quantities required, keys and footnotes to help in selecting correct parts.

6. Parts Supersession Information. Part numbers that have this history will be displayed in the parts list in the order that they were superseded (from oldest to newest). The superseded part numbers will be shown with a line through them.

Five periods in the PART NO. column (.) indicate that the part is either *Not Serviced Separately* or there is a reference to another figure. A figure reference is denoted by a pointing hand followed by a figure number in the DESCRIPTION column (☞ Figure 10)

7. Keys are used to show two or more similar assemblies, RH and LH assembly parts, etc. Select the appropriate key, "A", "B", "C", "D", or "E" and the corresponding quantity column to find the required parts. Two periods in the QTY column (..) indicate that the part is not used for that assembly.
8. Indented descriptions under the DESCRIPTION column are used to indicate assemblies and

subparts of assemblies. In the example on the previous page, *BEARING CAP* (Item 2) is a subpart of the major assembly *CASE*.

9. Quantities shown are for one assembly. The quantities of the subpart *BEARING CAP* are indicated as two and two. This means two per *CASE* assembly.
10. Serial number breaks identify the most current parts information and are shown as follows:
 - **S/N D001H2628**
Indicates last serial number affected.
 - S/N D001H2629** →
Indicates the first serial number affected.
 - S/N D001H1553 ↔ S/N D001H3291**
Indicates a range of serial numbers affected.

ECN numbers for pending serial numbers are shown to allow for the inclusion of the most current parts information and are identified as follows:

- **S/N Pending Per ECN 63243**
Indicates a pending serial number for last unit affected.
- S/N Pending Per ECN 63243** →
Indicates a pending serial number for first unit affected.

Date breaks are used to indicate when a change occurred but a subsequent serial number break was not recorded. This is an arbitrary date usually based on the ECN release and are identified as follows:

- **D/B 3/00**
Indicates that a change occurred sometime within that month for the item affected.
- D/B 3/00** →
Indicates that a change occurred sometime within that month for the item affected.
- D/B 3/00 ↔ D/B 4/00**
Indicates a range of production dates of the item affected.

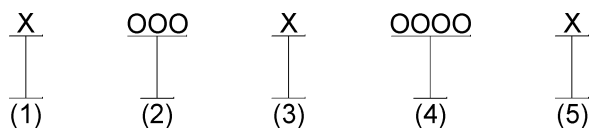
Contact your local dealer for Hyster lift trucks for serial number information for items with a ECN or a date break.

ORDERING PARTS:

When ordering replacement parts, give the unit serial number, part number, name of part and quantity required.

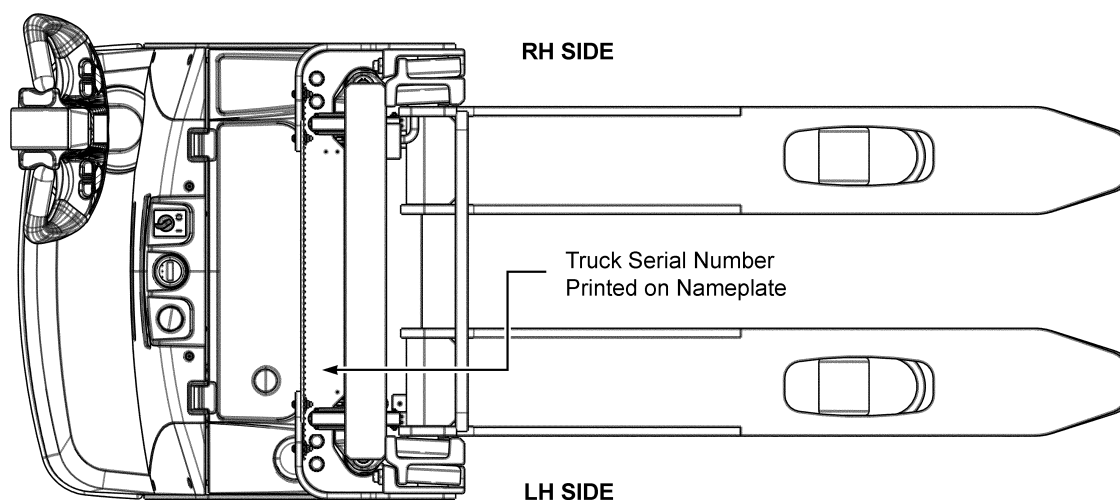
For any further information on parts, service, or ordering, consult your local dealer for Hyster lift trucks.

IDENTIFICATION OF TRUCK SERIAL NUMBERS



- (1) and (2) The first letter and three numbers indicate the truck unit code.
- (3) The second letter indicates the plant of manufacture.
- (4) The four or five digit number indicates the production sequence number of the truck.
- (5) The final letter indicates the year of manufacture starting with "A" in 1957, 1980, and 2003.
The letters "I", "O", and "Q" will not be used.

For pending serial numbers contact your local dealer for Hyster lift trucks.



BP990015

Parts referred to in this book as right or left hand parts are in accordance with the illustration above.

PUBLICATIONS

Every new lift truck is provided with an OPERATING MANUAL. This manual has information that the operator must know for inspection and operation of the lift truck.

SERVICE MAINTENANCE MANUALS, which provide service and overhaul procedures, and PARTS

MANUALS with complete replacement part identification, are available at a nominal fee from your Hyster dealer to assist users in effectively maintaining Hyster lift trucks in a safe, useful condition. See the list below for information on ordering specific publications for this model lift truck.

PUBLICATION	HYSTER PART NUMBER
Operating Manual.....	4148234
Service Manual (Complete).....	1668891
Parts Manual.....	860444

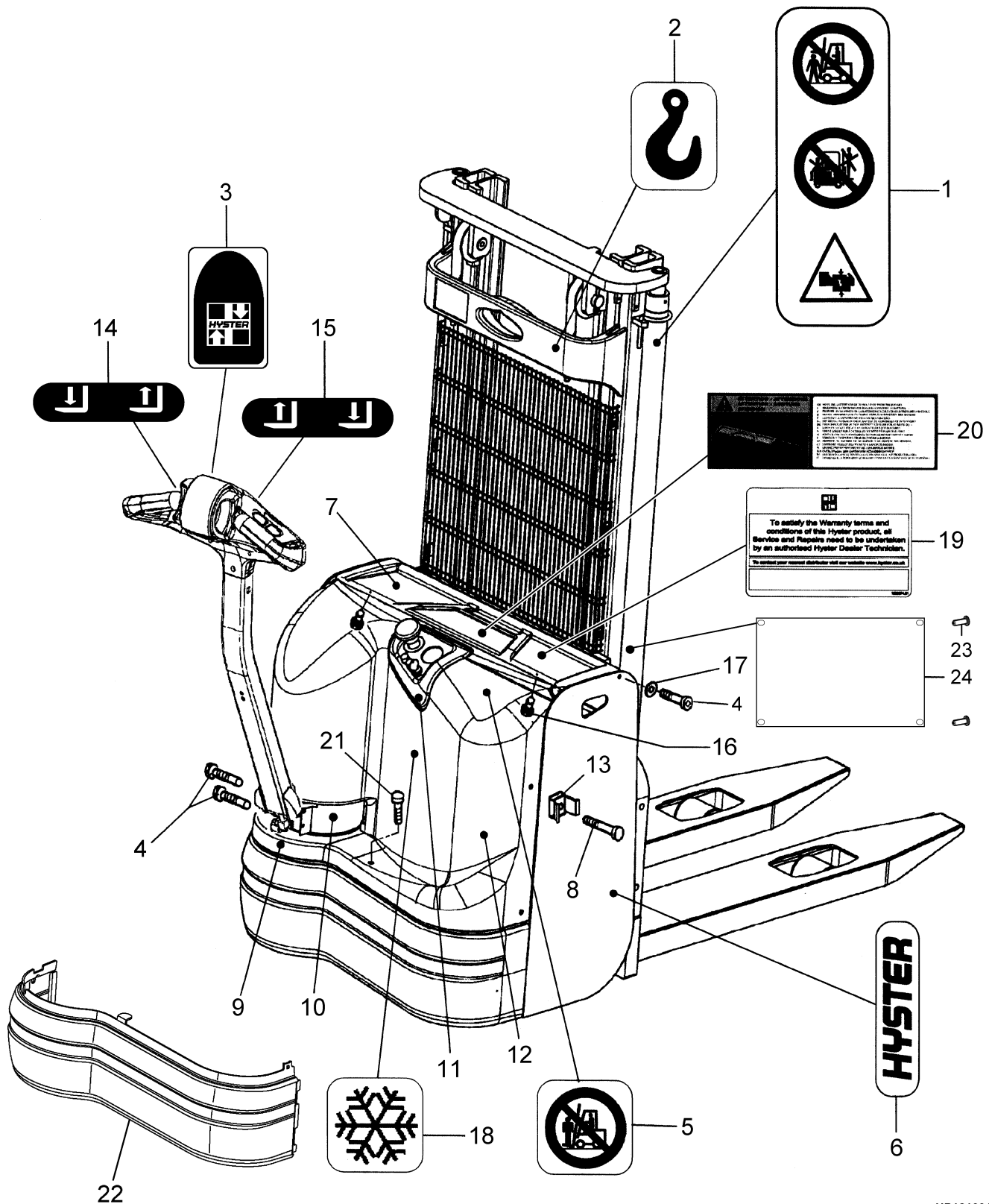
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FRAME/DRIVE UNIT

SECTION 1
FRAME/DRIVE UNIT

DESCRIPTION	PAGE
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COVERS AND ADHESIVE LABELS (FIGURE 1-1)

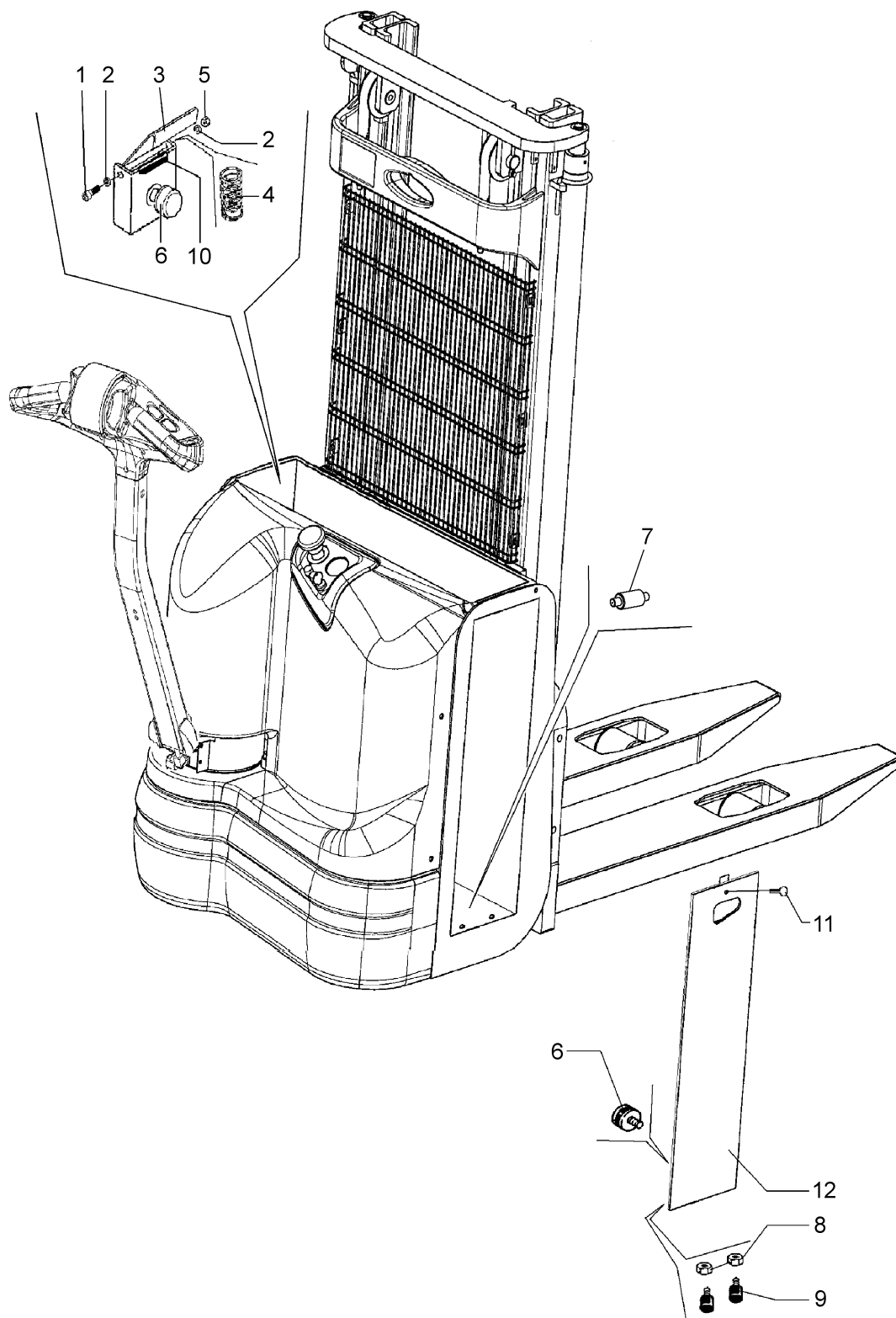


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COVERS AND ADHESIVE LABELS

ITEM	PART NO.	DESCRIPTION	QTY
1	2793761	Label (No Step/No Ride/Pinch Point)	2
2	2792746	Label (Lifting Hook)	1
3	4033244	Label (Tiller Handle Hyster)	1
4	293774	Capscrew	6
5	2792743	Label (No Riders)	1
6	2302743	Label (Hyster)	2
7	4151566	Cover	1
8	2794592	Screw	1
9	1476666	Cover , MDU Plate	1
10	2770386	Cover , Tiller Ring	1
11	4628132	Cover	1
12	1476664	Cover	1
13	1698418	Clip	4
14	4617882	Label (LH Switch Functions)	1
15	4614273	Label (RH Switch Functions)	1
16	2780616	Cushion	2
17	1466588	Spacer	2
18	2770775	Label (Snowflake)	1
19	1619955	Label (Warranty) (Brazilian)	1
20	1578418	Instruction Label	1
21	2793783	Screw	1
22	2770380	Bumper	1
23	222016	Rivet	4
24	4023318	Cover	1

BATTERY COMPARTMENT (FIGURE 1-2) LATERAL EXTRACTION



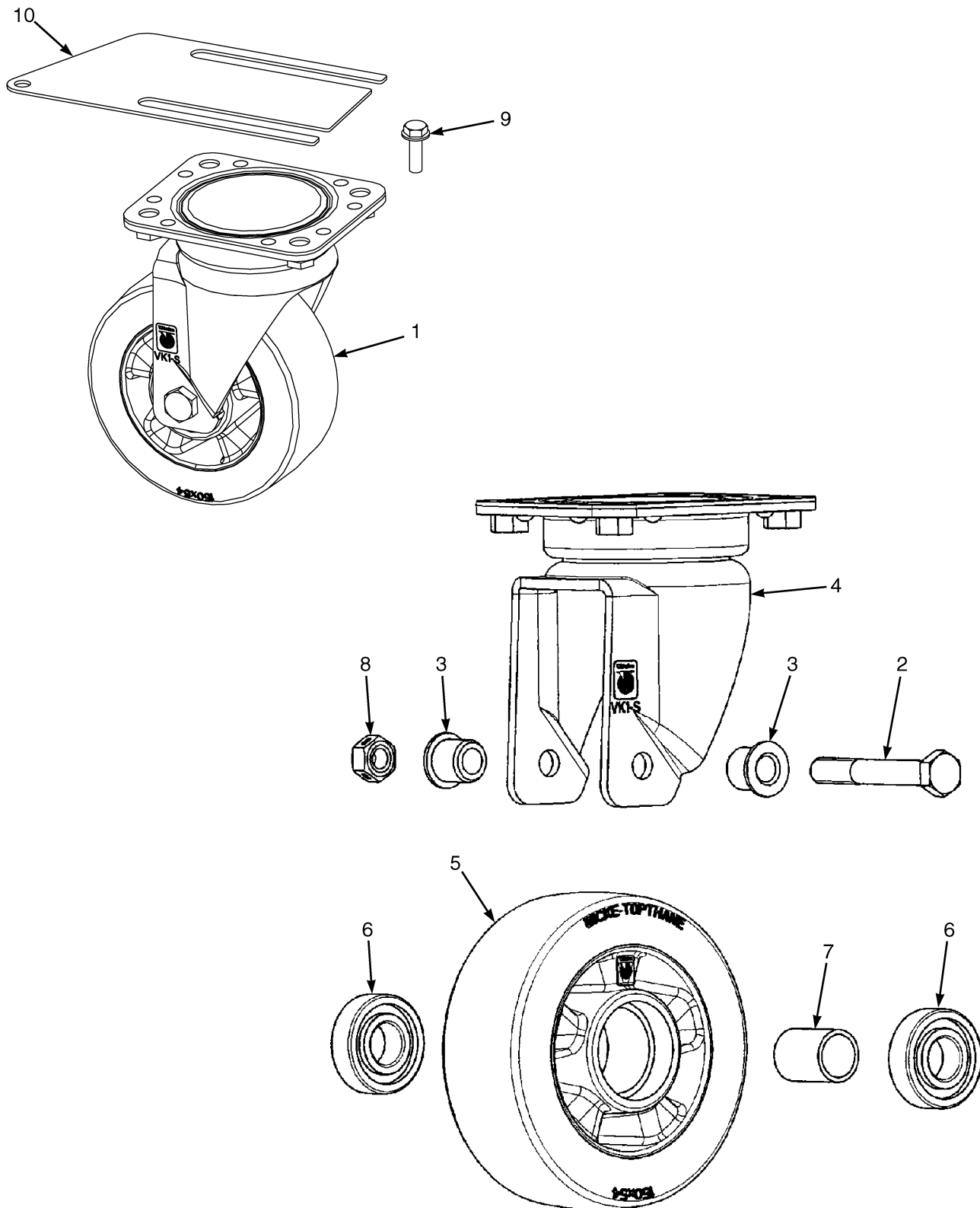
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**BATTERY COMPARTMENT
LATERAL EXTRACTION**

ITEM	PART NO.	DESCRIPTION	QTY
1	136749	Screw	1
2	320157	Washer	1
3	4152691	Hook	1
5	302400	Nut	1
6	2795502	Bumper	2
7	4055221	Roller	5
8	1512440		
	302399	Nut	2
9	2770378	Shock Absorber	2
11	4151231	Key	1
12	4151227	Cover	1
13	a 4151233	Lock (Not illustrated)	1

FOOTNOTES:**a** Not illustrated

SUPPORTS AND CASTER WHEELS (FIGURE 1-3)

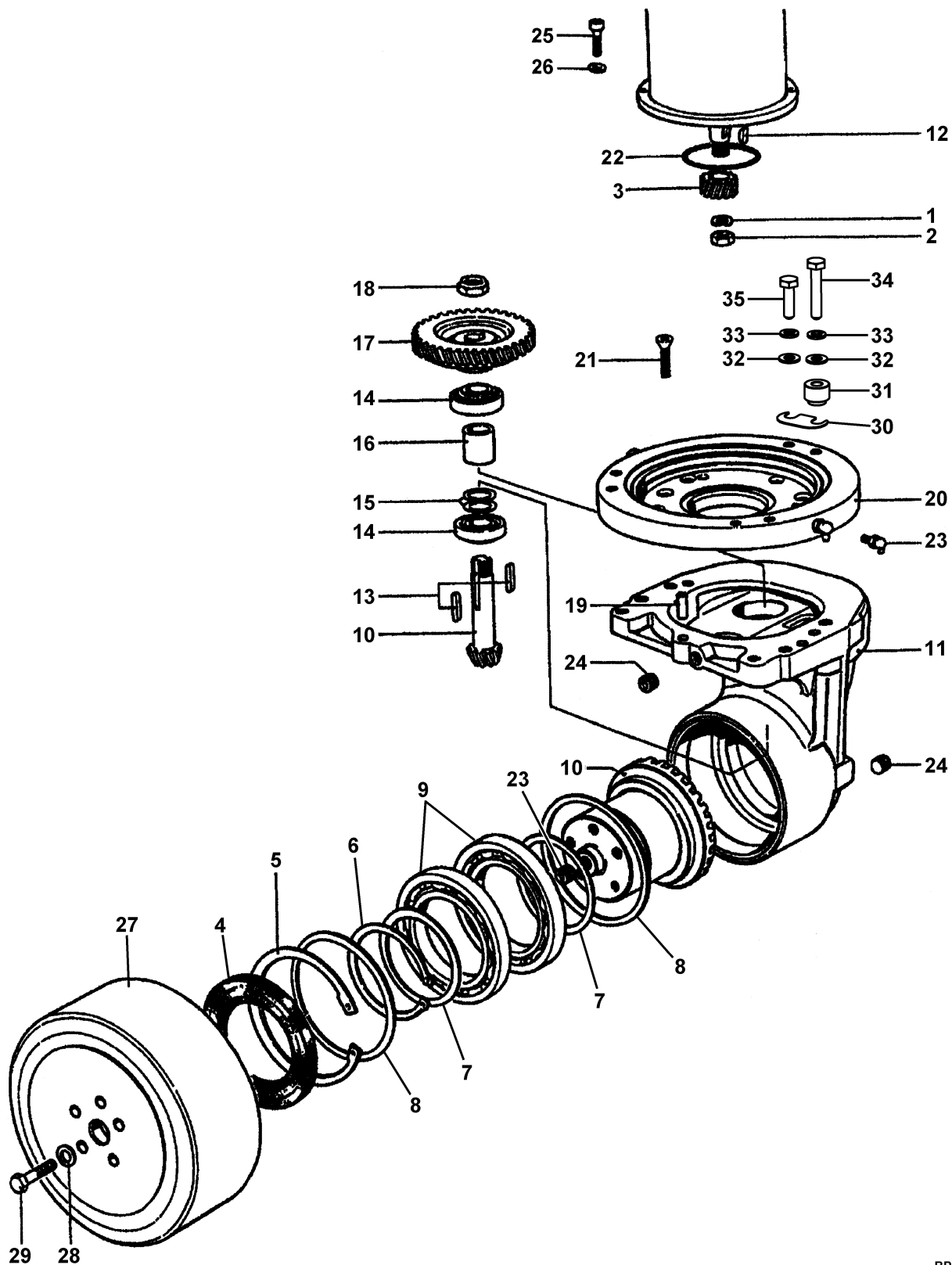


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SUPPORTS AND CASTER WHEELS

ITEM	PART NO.	DESCRIPTION	QTY
1	1694086	Caster Wheel Assembly	1
2	293656	Capscrew	1
3	1530390	Bushing	2
4	1694818	Swivel Fork	1
5	1694817	Wheel	1
6	4154346 128469	Ball Bearing	2
7	1694816	Axle Shaft	1
8	2045976	Locknut	1
9	2770825	Screw	4
10	1480077	Spacer (1 mm)	AR
10	2770354	Spacer (2 mm)	AR

TRACTION UNIT (FIGURE 1-4)

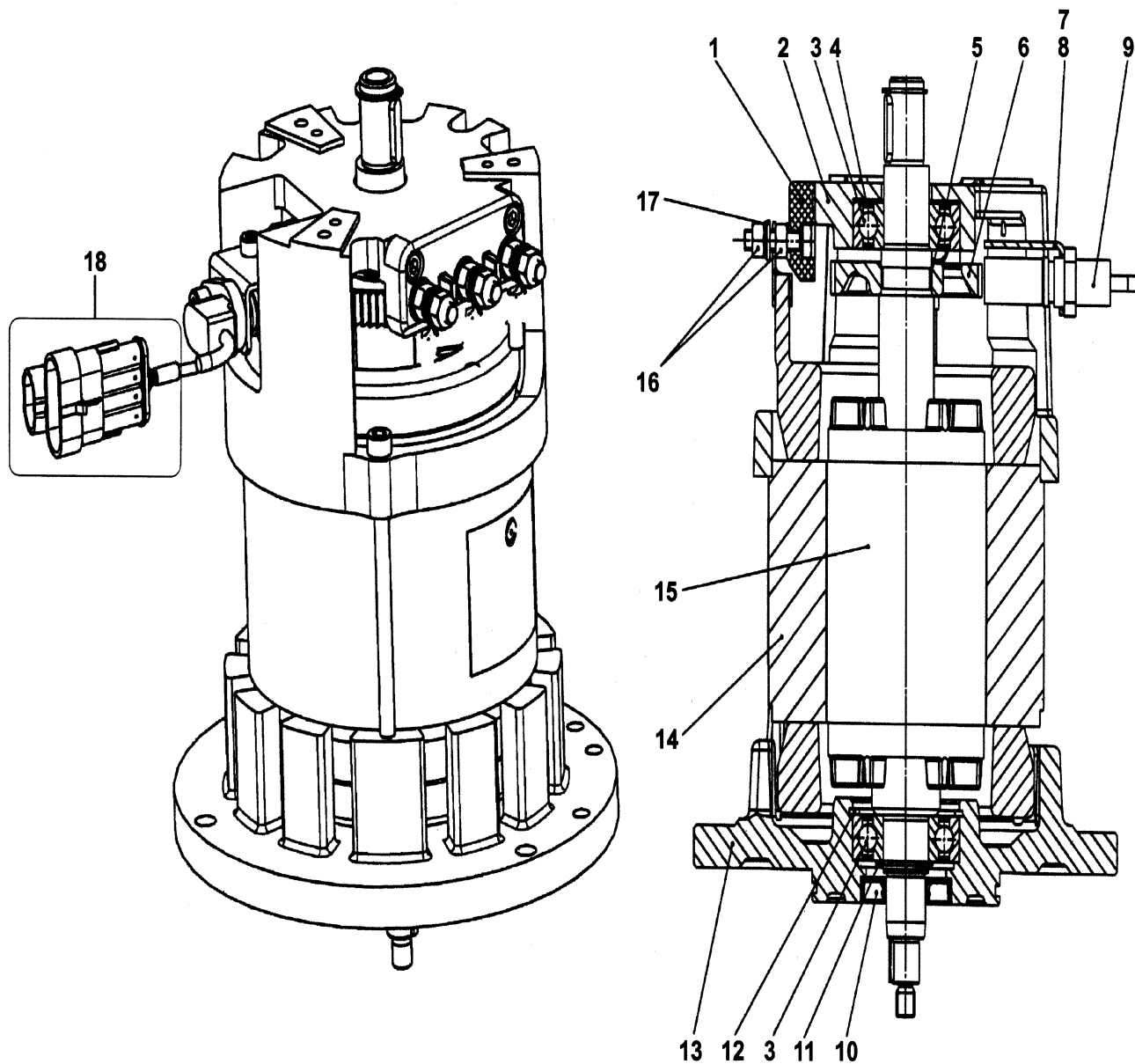


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TRACTION UNIT

ITEM	PART NO.	DESCRIPTION	QTY
	2060217	Transmission Assy	1
1	296418	Washer	1
2	239503	Nut	1
3	1539449	Gear	1
4	1470096	Seal	1
5	12981	Snap Ring	1
6	67551	Snap Ring	1
7	1470097	Shim	1
8	1470099	Shim	1
9	1470100	Bearing	2
10	1470148	Kit-Gear and Pinion	1
11	1539453	Transmission Housing	1
12	2794733	Key	1
13	2793258	Key	2
14	2792009	Bearing	2
15	2791933	Shim	1
16	2792105	Spacer	1
17	1539448	Gear	1
18	2794669	Nut	1
19	2792314	Screw	2
20	1470105	Bearing	1
21	296240	Capscrew	6
22	2791975	Gasket	1
23	1470108	Fitting	2
24	1470106	Drain Plug	3
25	296011	Capscrew	3
26	2792165	Screw	3
27	2770500	Drive Wheel (Tophane, 92Sh)	1
27	1491574	Drive Wheel (Rubber, 73Sh)	1
27	2770501	Drive Wheel (Non-Marking, 75Sh)	1
27	2770502	Drive Wheel (Gripped) (Cold Storage, 75Sh)	1
27	1532113	Drive Wheel (Antistatic Vulkollan 92Sh)	1
27	1539161	Drive Wheel (Vulkollan, 92Sh)	1
28	320161	Washer	5
29	4048844		
	292528	Capscrew	5
30	1580192	Shim	AR
31	2770489	Block	2
32	320159	Washer	6
33	388049		
	292682	Lockwasher	6
34	292518	Capscrew	2
35	287860	Capscrew	4

TRACTION MOTOR (FIGURE 1-5)

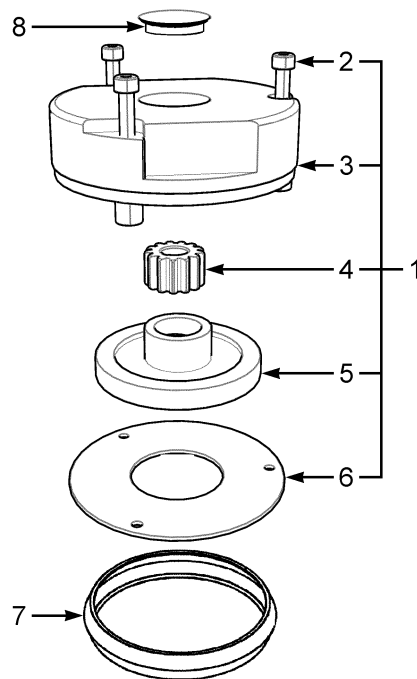


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TRACTION MOTOR

ITEM	PART NO.	DESCRIPTION	QTY
	2106374	Electric Motor Assembly	1
1	1700499	Terminal	1
2	1700501	End Cover	1
3	1470185	Bearing	2
4	1626085	Washer	1
5	1700507	Ring	1
6	1700505	Wheel	1
7	1700503	Support Plate	1
8	1700504	Spacer	1
9	1700502		
	4012949	Speed Sensor	1
10	1701031	Oil Seal	1
11	12929	Snap Ring	1
12	12928	Snap Ring	1
13	1700500	End Cover	1
14	1700498	Stator	1
15	1700506	Rotor	1
16	136687	Nut	6
17	388048	Disk	3
18	1591525	Sensor (Temperature)	1

ELECTROMAGNETIC BRAKE (FIGURE 1-6)



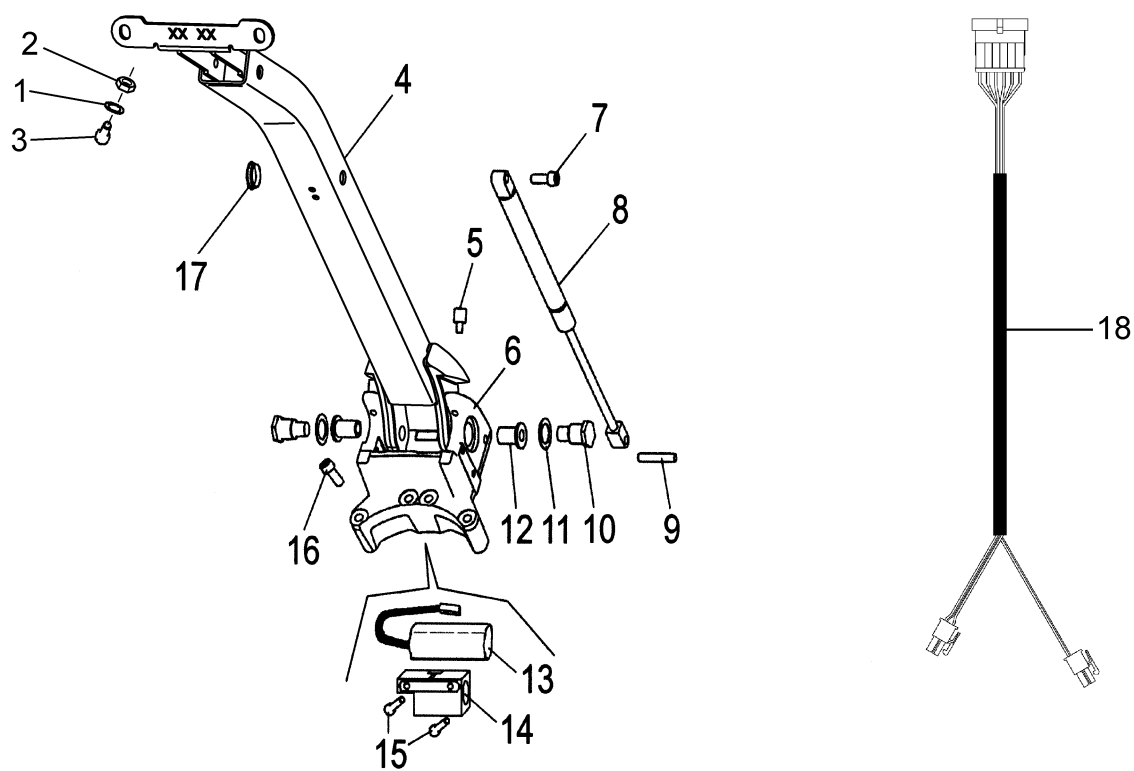
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ITEM	PART NO.	DESCRIPTION	QTY
1	4623719		
	4647352	Brake Assy	1
2	886889	Capscrew	3
3	4075556	Coil	1
4	1524569	Hub	1
5	4075554	Brake Disk	1
6	1682988	Flange	1
7	1682989	Band	1
8	2069534	Dust Cap	1

NOTES

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TILLER (FIGURE 1-7)



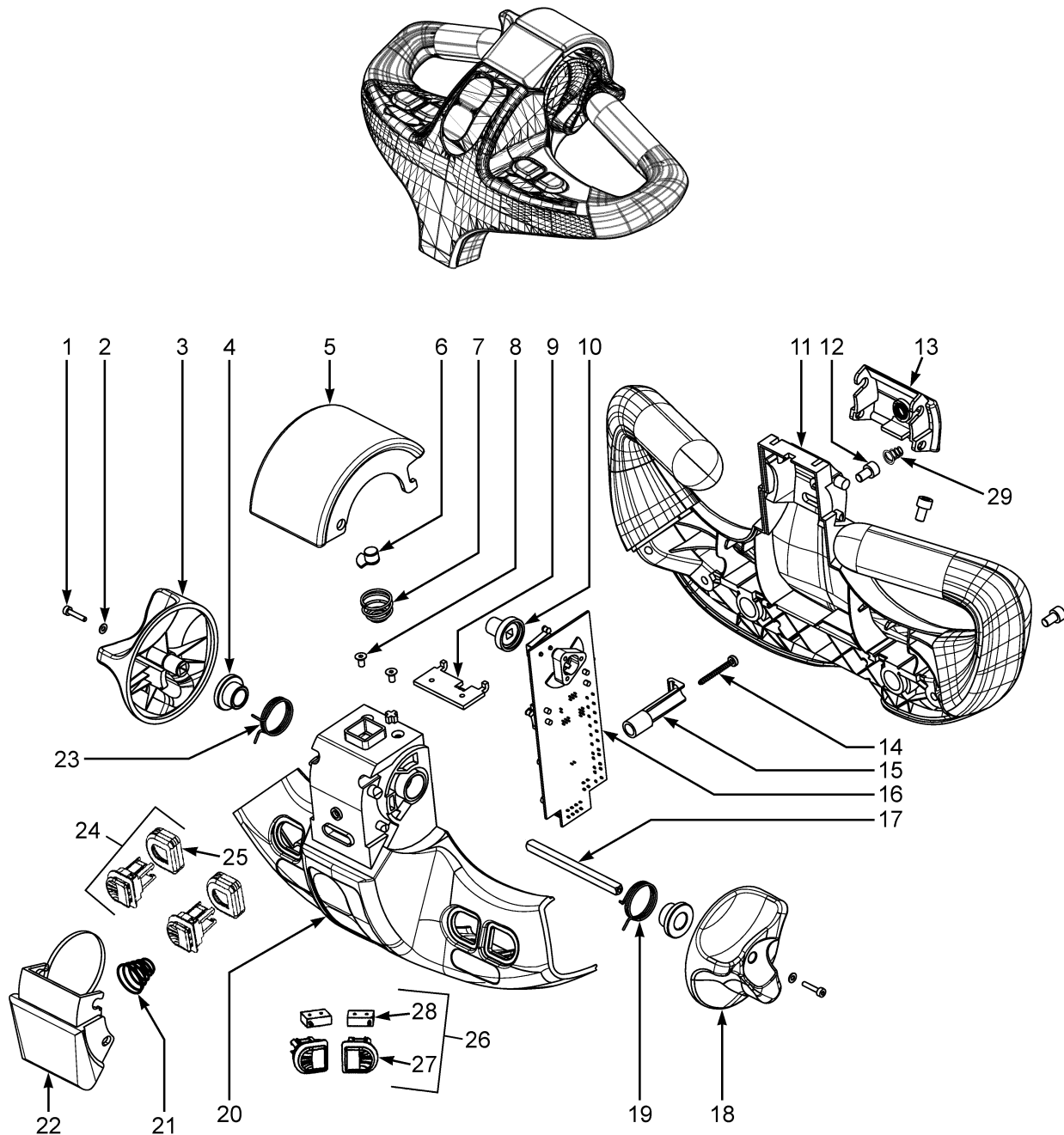
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TILLER

ITEM	PART NO.	DESCRIPTION	QTY
1	320159	Washer	2
2	296128	Nut	2
3	296027	Capscrew	2
4	4151377	Tiller Wldmt	1
5	2770378	Shock Absorber	2
6	2770353	Casting	1
7	2792378	Screw	1
8	1525904	Shock Absorber	1
9	2792295	Pin	1
10	2792053	Pin	2
11	2791931	Washer	2
12	2791987	Bushing	2
13	2038546		
	4601335	Sensor	1
14	2794114	Support	1
15	2795461	Screw	2
15	293727	Capscrew	2
15	a 320154	Washer	2
16	2770642	Screw	4
16	296017	Capscrew	4
16	a 320157	Washer	4
17	2790866	Plug	1
18	4630346	Wire Harness , Tiller	1

FOOTNOTES:**a** Not Illustrated

CONTROL BOX (FIGURE 1-8) PROPORTIONAL VERSION



BP220257

**CONTROL BOX
PROPORTIONAL VERSION**

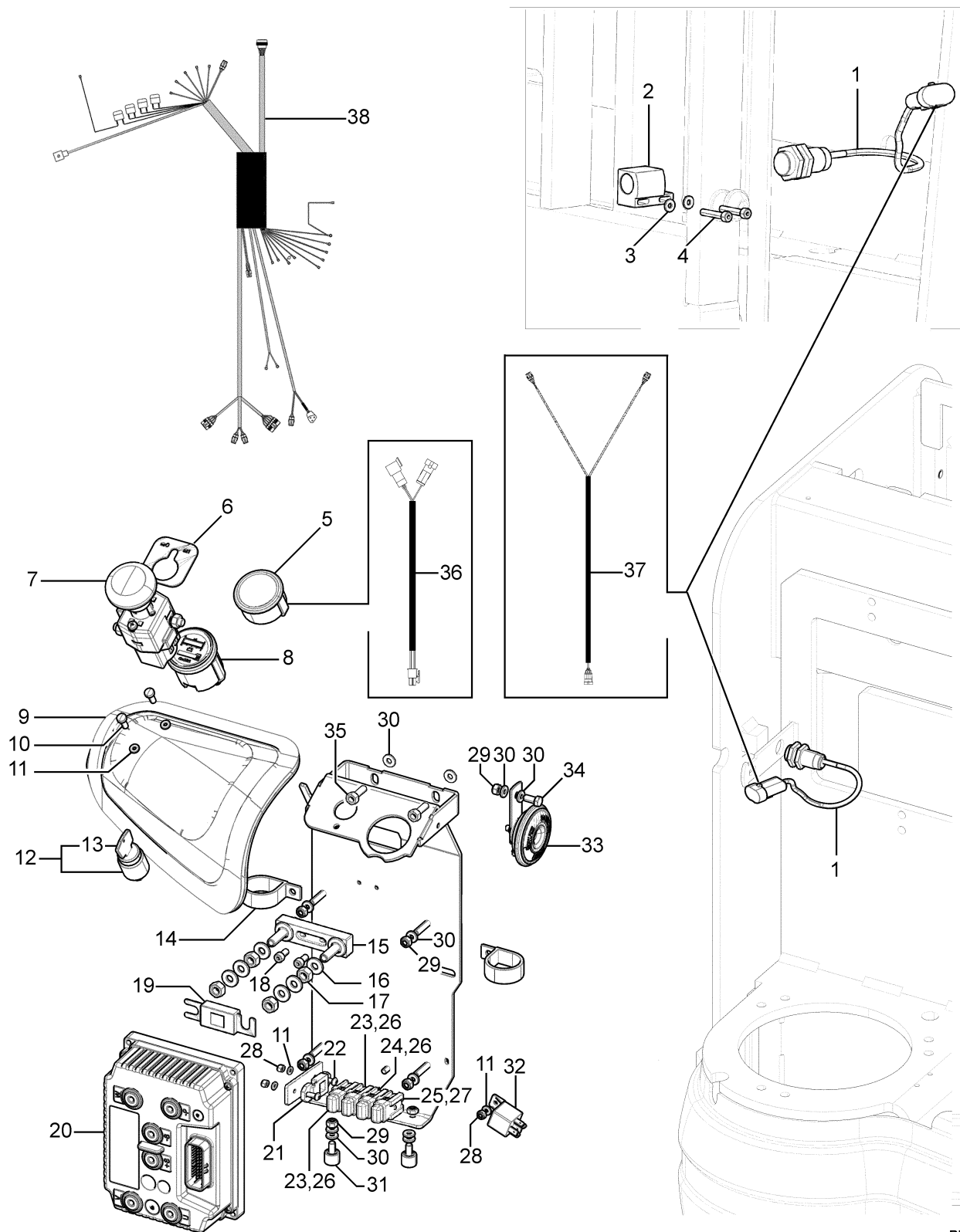
ITEM	PART NO.	DESCRIPTION	QTY
	4084426	Tiller Head	1
1	1499660	Capscrew	2
2	320152	Washer	2
3	4013194	Accelerator Control Knob (RH Butterfly)	1
4	1466595	Bushing	2
5	4023859		
	1467779	Cover (Belly Switch)	1
6	1660875	Locking Lever (Belly Spring)	1
7	1531075	Spring (Cover, Belly)	1
8	296192	Capscrew	2
9	1706800	Plate	1
10	4036580	Bushing	1
11	2111838	Tiller Head Assembly (Lower Half) (Composite)	1
12	293761	Capscrew	3
13	4013196	Cover , Rear (For Push Button)	1
14	2097314	Screw	1
15	1652960	Support Clip	1
16	4036431	Printed Circuit Board (MULTICAN)	1
17	4023864		
	1466935	Camshaft , Direction Control	1
18	4013193	Accelerator Control Knob	1
19	1594131	Spring , Directional Control Return	1
20	4611400	Tiller Head (Upper Half)	1
21	1466606	Spring (Horn/Snail)	1
22	4011997	Horn Button Cover	1
23	1466607	Spring , Directional Control Return	1
24	4046682	Push Button Switch Assy (Proportional)	2
25	1581353	Rubber Cover (Proportional Version)	1
26	4011993	Push Button Switch Assembly (On-Off)	2
27	4036439	Push Button Switch (On-Off)	1
28	1498102	Microswitch (On-Off, Molex Connector)	1
29	2043556	Spring (Quick Pick)	1

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

SECTION 2
ELECTRICAL SYSTEM

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POWER CABLES	2-4

ELECTRICAL ARRANGEMENT (FIGURE 2-1)

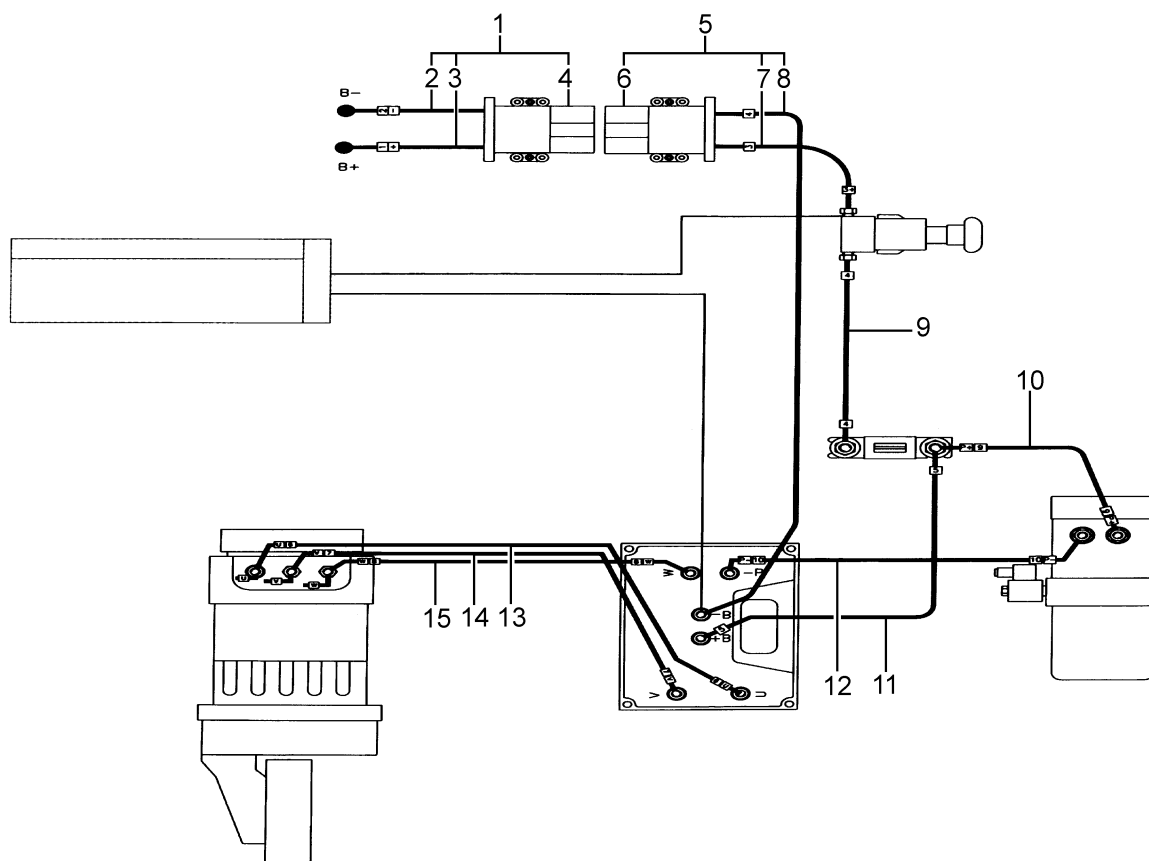


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ELECTRICAL ARRANGEMENT

ITEM	PART NO.	DESCRIPTION	QTY
7	1693742	Switch (Emergency Disconnect)	1
8	2790487	BDI And Hourmeter	1
9	4628132	Cover	1
10	296509	Screw	2
11	320155	Washer	5
12	1699776 4038296	Ignition Switch	1
13	4041497	Key	1
14	339530	Clamp	2
15	4096945	Isolator	1
16	320157	Washer	6
17	239503	Nut	4
18	293742	Screw	2
19	1648884	Fuse (300 Amp)	1
20	2108634	Controller	1
21	1588252	Support	1
22	296153	Capscrew	2
23	1337828	Fuse (3 Amp)	2
24	370646	Fuse (5 Amp)	1
25	370647	Fuse (10 Amp)	1
26	1590162	Fuse Holder	3
27	1589277	Fuse Holder (10 Amp)	1
28	302398	Locknut	3
29	1512440 302399	Nut	7
30	4093489 320156	Washer	12
31	2770378	Shock Absorber	2
32	4641976	Relay (Mini SPDT WP with Diode)	1
33	1692973	Horn	1
34	292500	Capscrew	1
35	292501	Capscrew	2
38	2108354	Wire Harness	1

POWER CABLES (FIGURE 2-2)



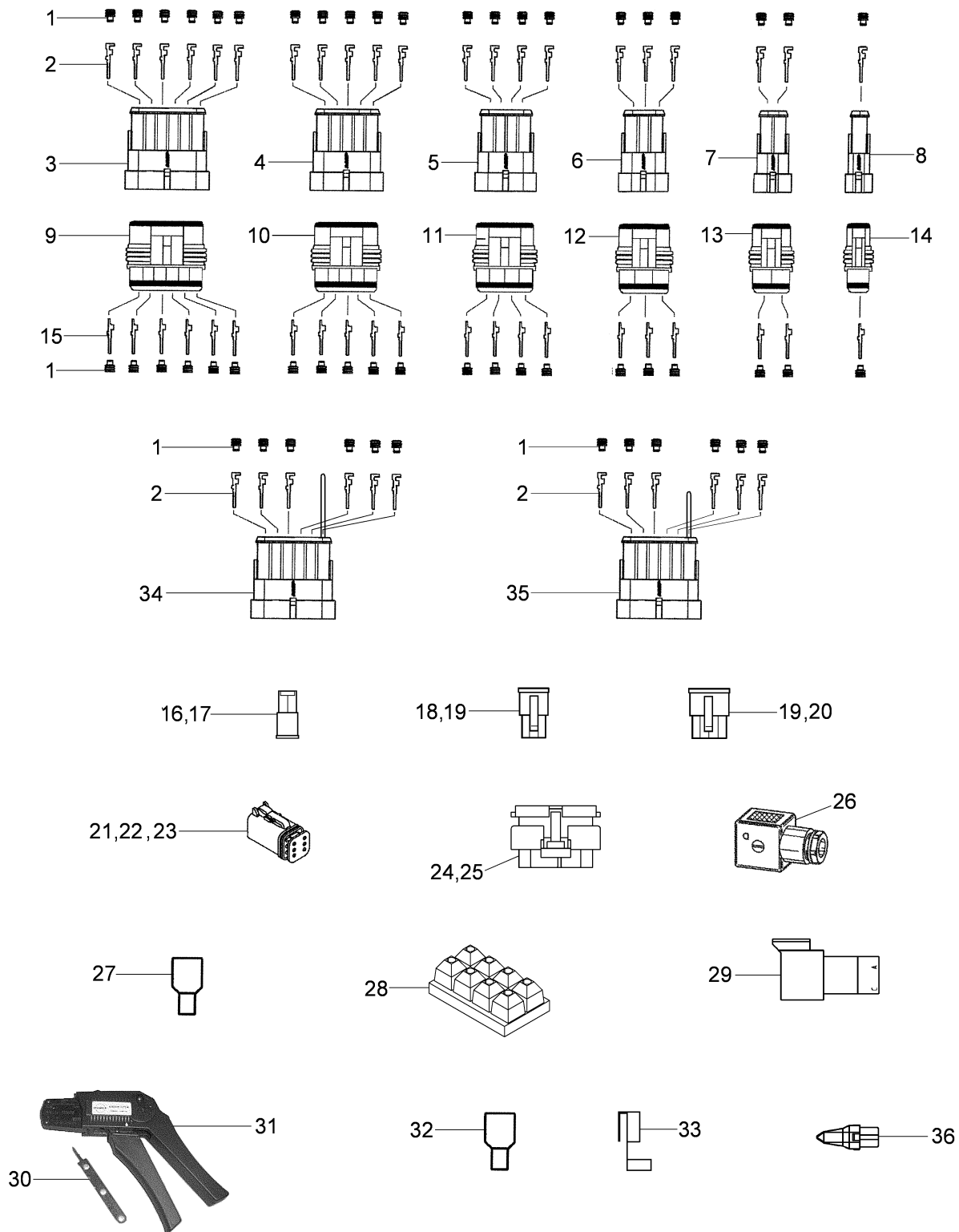
BP081234

ITEM	PART NO.	DESCRIPTION	QTY
1	1532285	Battery Connector	1
2	2059400	Battery Cable (-)	1
3	2059399	Battery Cable (+)	1
4	1532284	Battery Connector Housing	1
5	1532287	Battery Connector and Cable Assembly	1
6	1532286	Battery Connector Housing	1
7	2059401	Battery Cable (+)	1
8	2059402	Battery Cable (-)	1
9	2108492	Power Cable	1
10	2108282	Power Cable	1
11	2108284	Power Cable	1
12	2108283	Power Cable	1
13	2108279	Power Cable	1
14	2108280	Power Cable	1
15	2108281	Power Cable	1

NOTES

[illegible]

CONNECTORS, CONTACTS, TOOLS (FIGURE 2-3)



BP081471

CONNECTORS, CONTACTS, TOOLS

ITEM	PART NO.	DESCRIPTION	QTY
1	2060555	Wire Seal (Yellow)	AR
2	2787350	Pin Terminal	AR
3	1608759	Electrical Connector (6 Way)	AR
4	2787339	Terminal (5 Way)	AR
5	1591532		
	2787341	Connector (4 Way)	AR
6	2795583	Connector (3 Way)	AR
7	2787345	Connector (2 Way)	AR
8	2067918	Connector (1 Way)	AR
9	1608758	Electrical Connector (6 Way)	AR
10	2787340	Connector (5 Way)	AR
11	1508211	Connector (4 Way)	AR
12	1547629	Connector (3 Way)	AR
13	1508206	Connector (2 Way)	AR
14	2067917	Connector (1 Way)	AR
15	1508232	Terminal (Pin)	AR
16	2059812	Connector (4 Way)	AR
17	2302852	Pin	AR
18	2048593	Connector (6 Way)	AR
19	2309579	Socket	AR
20	1369409	Connector (8 Way)	AR
21	1330420	Connector	AR
22	1330438	Terminal (Socket)	AR
23	1610944	Lock	AR
24	2056243	Connector (35-Way)	AR
25	1561019		
	2035276	Connector (Socket Female)	AR
26	2060614	Connector	AR
27	1520400	Blade Terminal (Red - Sec. 1,5mm - Male)	AR
28	2048835	Insulator	AR
29	1330410	Connector	AR
30	1471485	Extractor	1
31	1471487	Pliers	1
32	1656304	Terminal (Blade)	AR
33	4002239	Terminal	AR
34	4020725	Connector (With Bridge)	AR
35	4020726	Connector (With Bridge)	AR
36	1532663	Connector (Secondary Lock)	AR

[illegible]

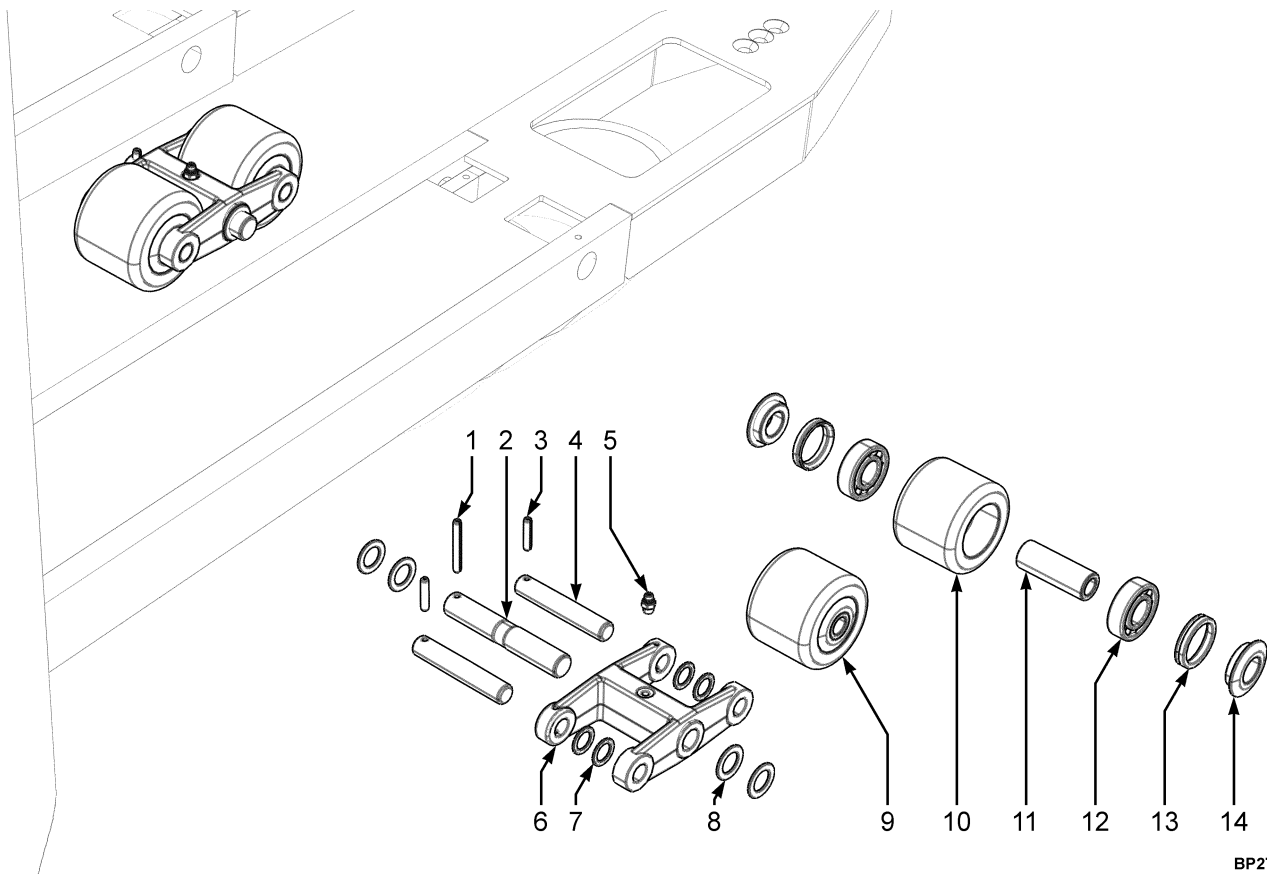
3

LOAD WHEEL

**SECTION 3
LOAD WHEEL**

DESCRIPTION	PAGE
LOAD WHEELS	
TANDEM	3-2

LOAD WHEELS (FIGURE 3-1) TANDEM



BP270115

ITEM	PART NO.	DESCRIPTION	QTY
1	302456	Roll Pin	2
2	2770370	Pin	2
3	302452	Roll Pin	2
4	2792265	Pin	4
5	2791942	Grease Fitting	2
6	2770656	Rocker Arm	2
7	2792219	Washer	8
8	2791931	Washer	8
9	1531785	Roller Assembly	4
10	1531787	Roller	1
11	1531788	Bushing	1
12	194723	Ball Bearing	2
13	3041219	Seal	2
14	1530189	Bushing	2

SECTION 4
HYDRAULIC SYSTEM

DESCRIPTION	PAGE
HYDRAULIC ARRANGEMENT	4-2
HYDRAULIC POWER PACK	4-4

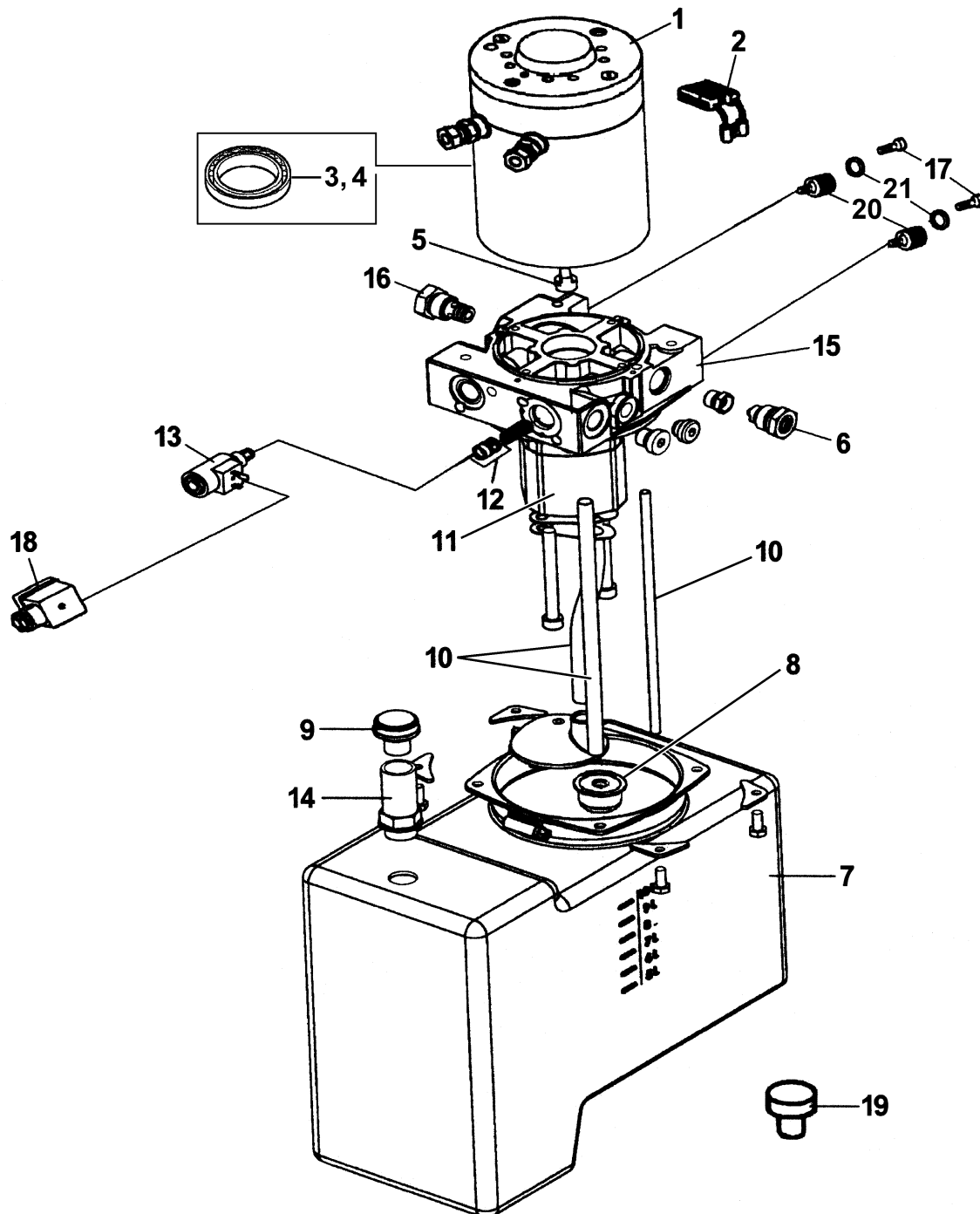
2 stage LFL



HYDRAULIC ARRANGEMENT**KEY:****A: 2 STG LFL****B: 3 STG FFL**

ITEM	PART NO.	DESCRIPTION	QTY	
			A	B
1	2798663	Fitting	2	3
2	2798661	Fitting	2	1
3	2770396	Hydraulic Tube	1	1
4	2770686	Tube Assy	1	1
5	14560	Fitting	..	1
6	2770690	Tube	..	1
7	16447	Fitting	..	2
8	2770681	Hose (Lift Height=4625mm)	..	1
8	1482608	Hose (Lift Height=5395mm)	..	1
9	17319	Fitting	..	1
10	17218	Fitting	..	1
11	2770689	Tube (Lift Height=4625mm)	..	1
11	1482587	Tube (Lift Height=5395mm)	..	1

HYDRAULIC POWER PACK (FIGURE 4-2)



BP150310

HYDRAULIC POWER PACK

ITEM	PART NO.	DESCRIPTION	QTY
	2770677	Pump And Motor Assembly	1
1	2770785	Motor Assy	1
2	2770786	Kit-Brush	1
3	1150086	Bearing	1
4	4021991	Bearing	1
5	1495150	Coupling	1
6	1470170	Relief Valve	2
7	1600294	Reservoir	1
8	355859	Strainer	1
9	1503998	Hyd Oil Fill Cap	1
10	1600793	Tube Assy	1
11	1470171	Pump	1
12	1606701	Flow Valve	1
13	2770675	Valve , Proportional	1
14	1503997	Tube Assy	1
15	1470169	Distributor	1
16	355866	Valve	1
17	287854	Capscrew	2
19	2770679	Support , Rubber	1
20	2791974		
	4004119	Vibration Damper	2
21	320159	Washer	2
	1470172	Hydraulic Tank Kit (Includes Items 7-10, 14)	1

[illegible]

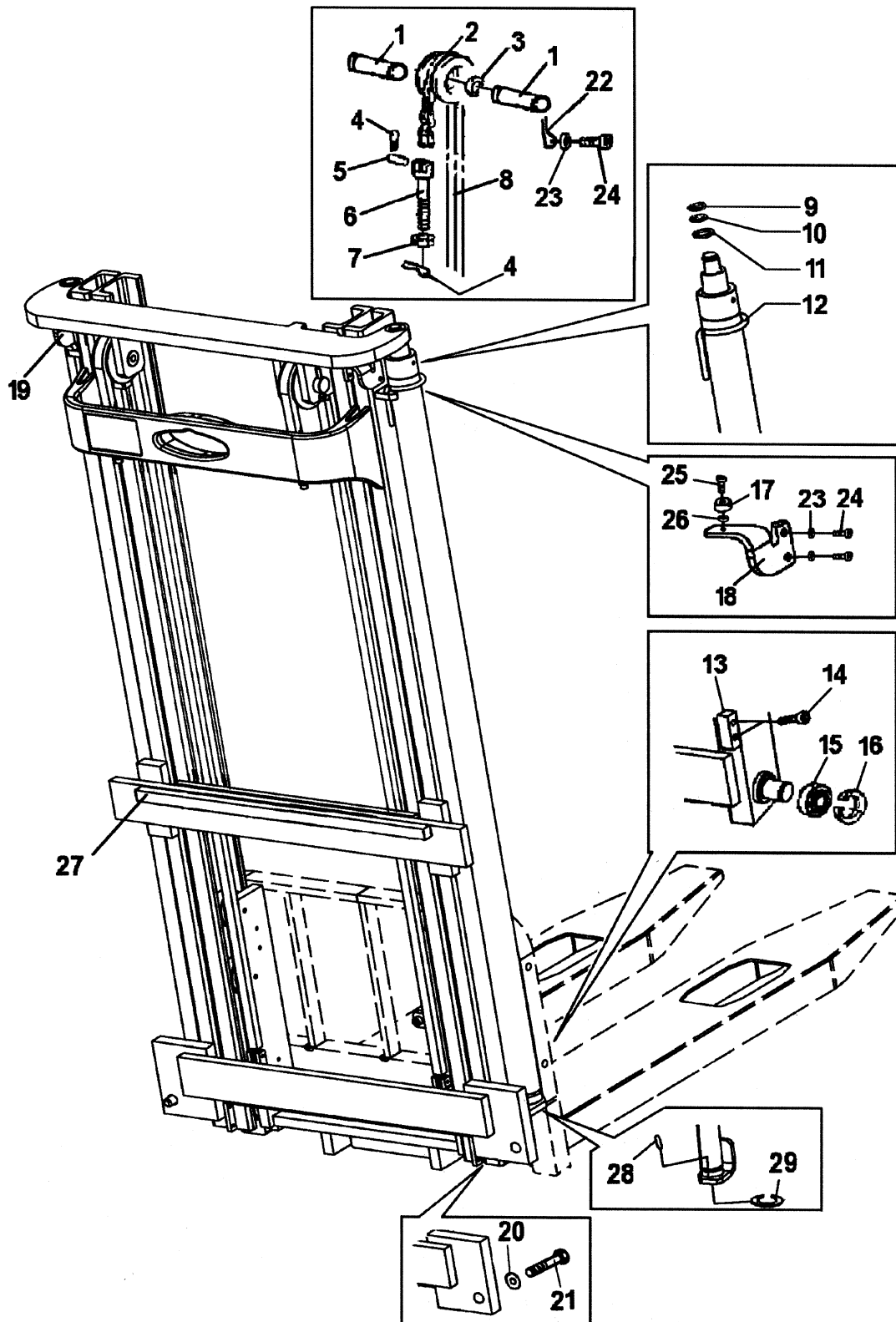
5

MAST-TWO STAGE LIMITED FREE-LIFT

SECTION 5
MAST-TWO STAGE LIMITED FREE-LIFT

DESCRIPTION	PAGE
HYDRAULIC CYLINDERS	
TWO STAGE LIMITED FREE-LIFT MAST	5-4
MAST	
TWO STAGE LIMITED FREE-LIFT	5-2
WIRE NET	
TWO STAGE LIMITED FREE-LIFT MAST	5-6

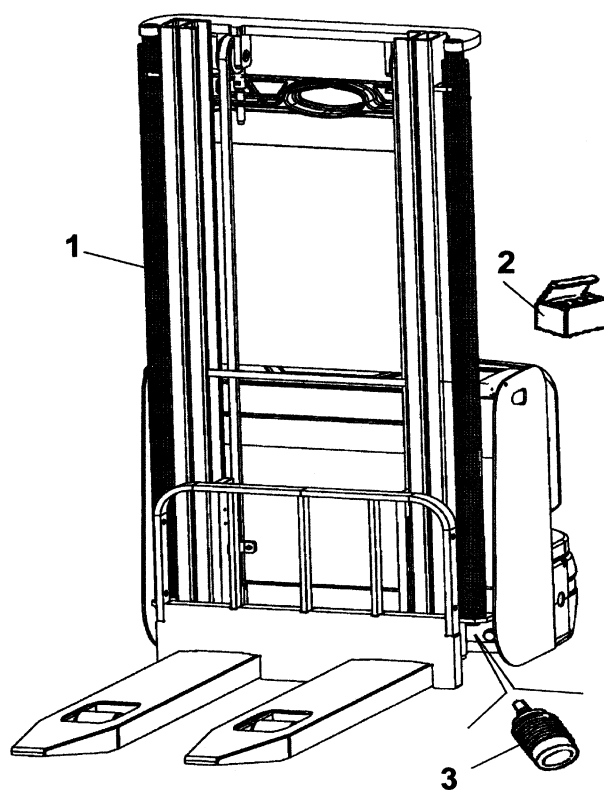
MAST (FIGURE 5-1) TWO STAGE LIMITED FREE-LIFT



BP170400

MAST
TWO STAGE LIMITED FREE-LIFT

ITEM	PART NO.	DESCRIPTION	QTY
1	1465597	Pin	2
2	2790663	Pulley	2
3	2793383	Washer	2
4	15201	Cotter Pin	12
5	2793850	Pin	4
6	2793851	Anchor	2
7	302404	Locknut	2
8	1468061	Chain (Lift Height=4168mm)	2
9	131357	Snap Ring	2
10	4020900		
	320169	Washer	2
11	2045629	Seal	2
12	2770157	Brace	2
13	2795456	Wear Plate (0.12 in.) (Not Used)	AR
14	296215	Capscrew (Not Used)	AR
15	1363272	Load Roller ("B") (3.24 in.)	10
16	58901	Snap Ring	10
17	2794101	Damper	2
18	2770276	Bracket (RH)	1
19	2770277	Bracket (LH)	1
20	292685	Lockwasher	2
21	296075	Capscrew	2
22	15262	Cotter Pin	2
23	292680	Washer	6
24	293774	Capscrew	6
25	293796	Capscrew	2
26	292680	Washer	AR
27	1497753	Foam	1
28	310433	Pin	2
29	304522	Snap Ring	2

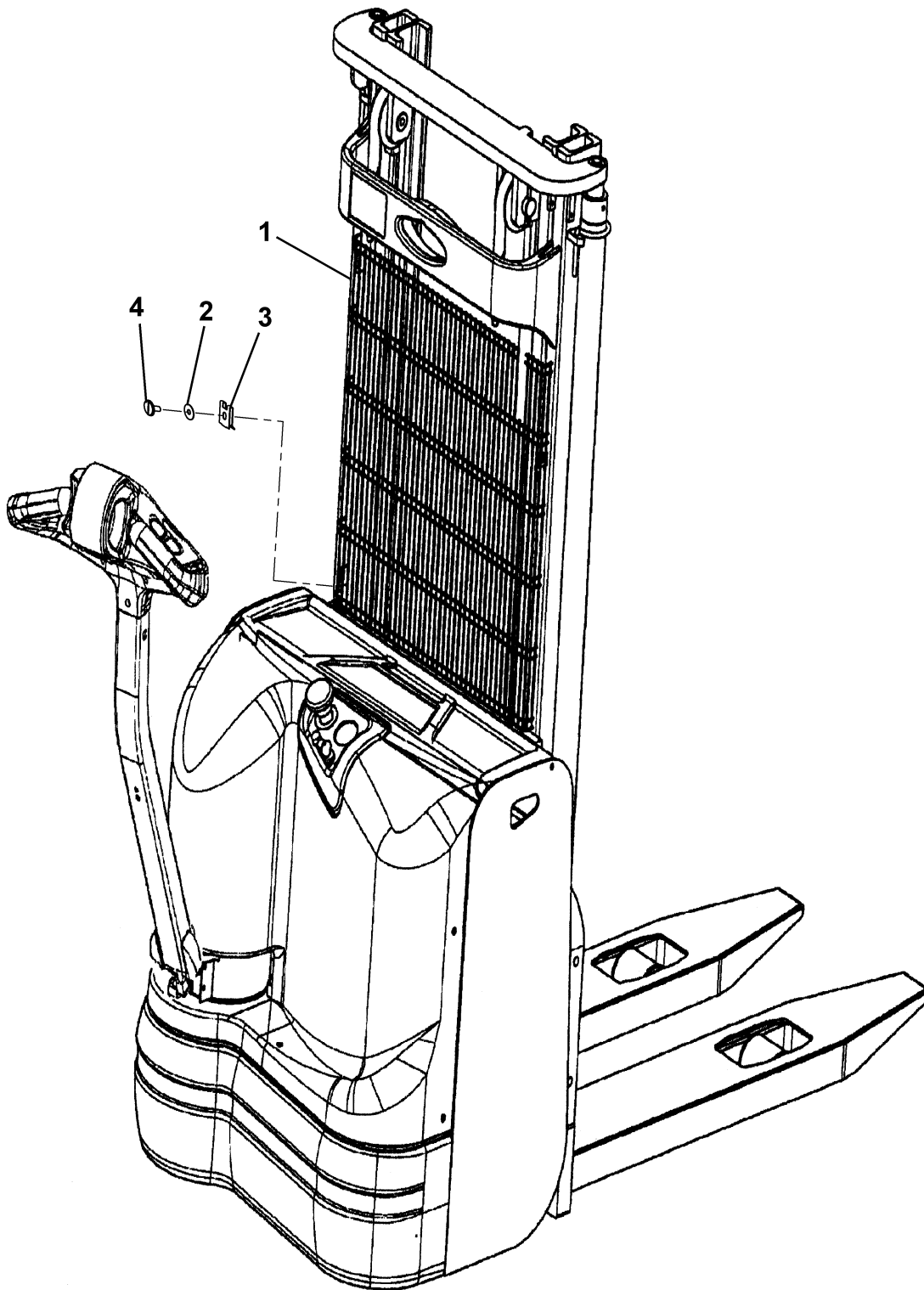
HYDRAULIC CYLINDERS (FIGURE 5-2)
TWO STAGE LIMITED FREE-LIFT MAST

BP170540

ITEM	PART NO.	DESCRIPTION	QTY
1	2039183	Cylinder Assy (Lift Height=4168mm)	2
2	2041531	Kit-Seal	1
3	893268	Flow Valve	2

NOTES

[illegible]

**WIRE NET (FIGURE 5-3)
TWO STAGE LIMITED FREE-LIFT MAST**

BP290133

WIRE NET
TWO STAGE LIMITED FREE-LIFT MAST

ITEM	PART NO.	DESCRIPTION	QTY
1	1482417	Guard (Lift Height=4168mm)	1
2	1608780	Washer	2
3	1585574	Nut , Clip	2
4	1587878	Screw	2

[illegible]

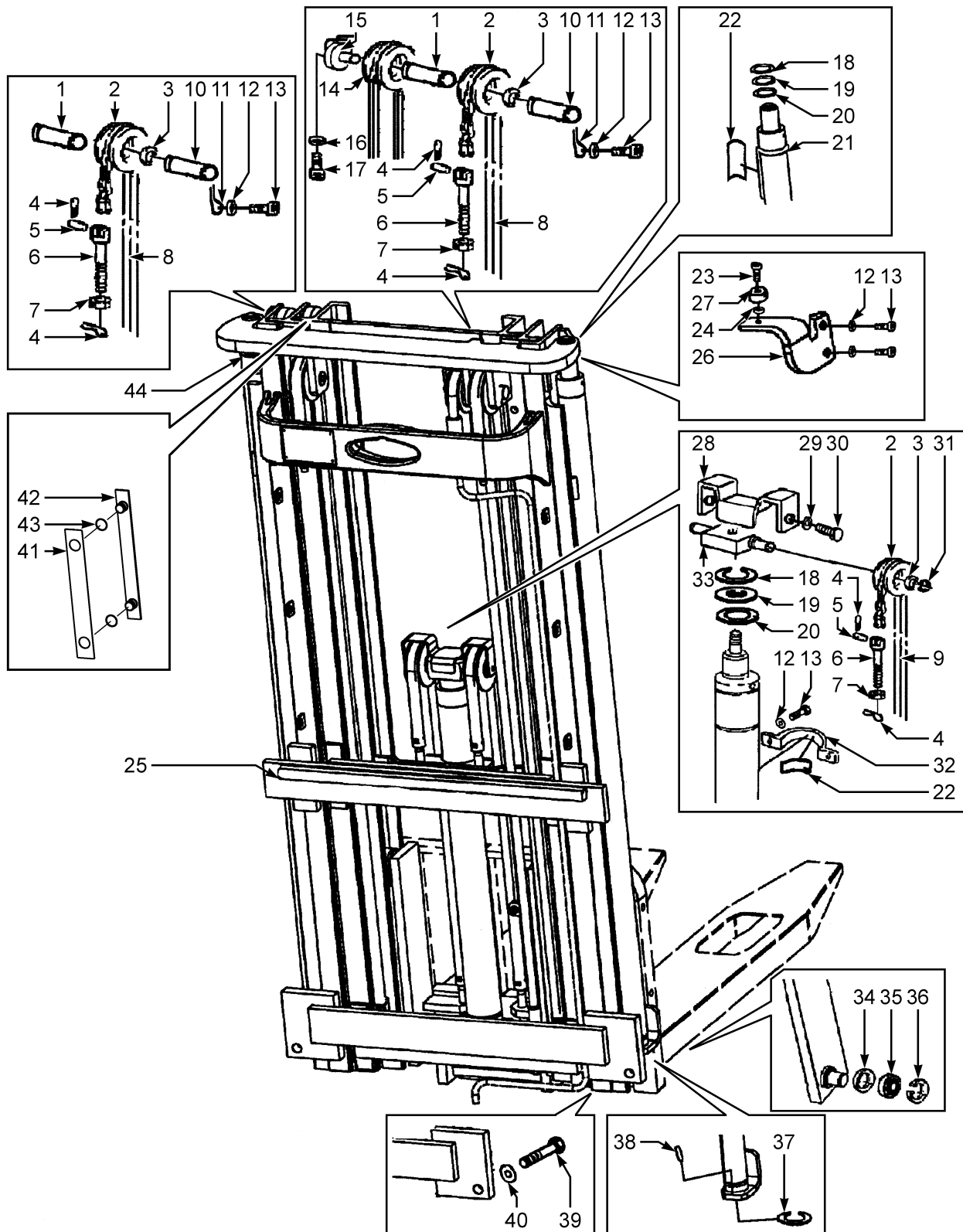
6

MAST-THREE STAGE FULL FREE-LIFT

SECTION 6
MAST-THREE STAGE FULL FREE-LIFT

DESCRIPTION	PAGE
HYDRAULIC CYLINDERS	
THREE STAGE FULL FREE-LIFT MAST	6-4
MAST	
THREE STAGE FULL FREE- LIFT	6-2
WIRE NET	
THREE STAGE MAST	6-6

MAST (FIGURE 6-1) THREE STAGE FULL FREE- LIFT

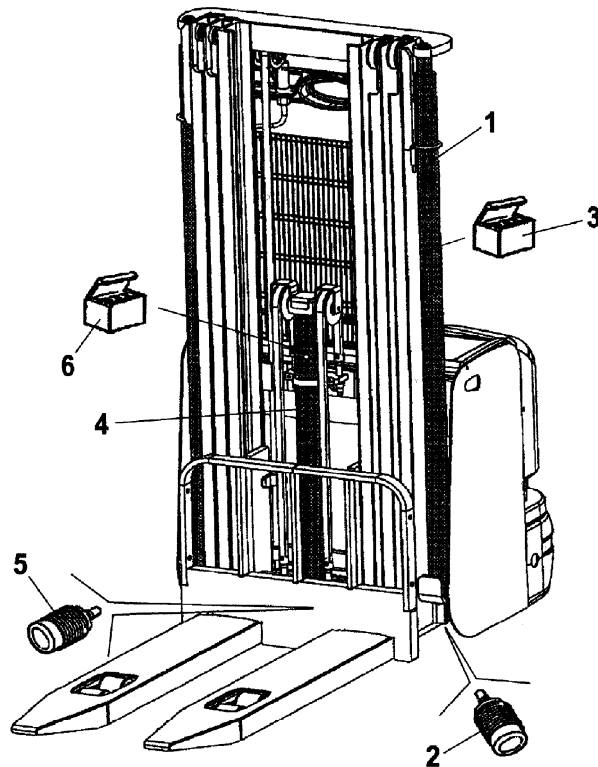


BP290206

MAST
THREE STAGE FULL FREE- LIFT

ITEM	PART NO.	DESCRIPTION	QTY
1	1465597	Pin	2
2	2790663	Pulley	4
3	2793383	Washer	10
4	15201	Cotter Pin	24
5	2793850	Pin	8
6	2793851	Anchor	6
7	302404	Locknut	6
8	1468058	Chain (Lift Height=4625mm)	2
8	1468060	Chain (Lift Height=5395mm)	2
9	1468053	Chain (Lift Height=4625mm)	2
9	1468055	Chain (Lift Height=5395mm)	2
10	1465598	Pin	1
11	15262	Cotter Pin	2
12	292680	Washer	8
13	293774	Capscrew	8
14	1343989	Hose Sheave	1
15	4034229	Stubshaft	1
16	292683	Lockwasher	1
17	292533	Capscrew	1
18	131357	Snap Ring	3
19	4020900		
	320169	Washer	AR
20	2791951	Seal	3
21	2770157	Brace	2
22	2795455	Rubber Pad	3
23	293796	Capscrew	2
24	292680	Washer	AR
25	1497753	Foam (Not Used)	AR
26	2770276	Bracket (RH)	1
27	2794101	Damper	2
28	4151710	Cover	2
29	292681	Lockwasher	2
30	287851	Capscrew	2
31	1692276		
	58908	Snap Ring	2
32	1484420	Strap	1
33	2770149	Sheave	1
34	129387	Shim (0.020 in.)	AR
35	1363272	Load Roller ("B") (3.24 in.)	12
36	58901	Snap Ring	12
37	304522	Snap Ring	3
38	310433	Pin	3
39	296075	Capscrew	2
40	292685	Lockwasher	2
41	131716	Shim (0.020 in.)	AR
42	215212	Wear Strip	4
43	104506	O-Ring	4
44	2770277	Bracket (LH)	1

HYDRAULIC CYLINDERS (FIGURE 6-2) THREE STAGE FULL FREE-LIFT MAST



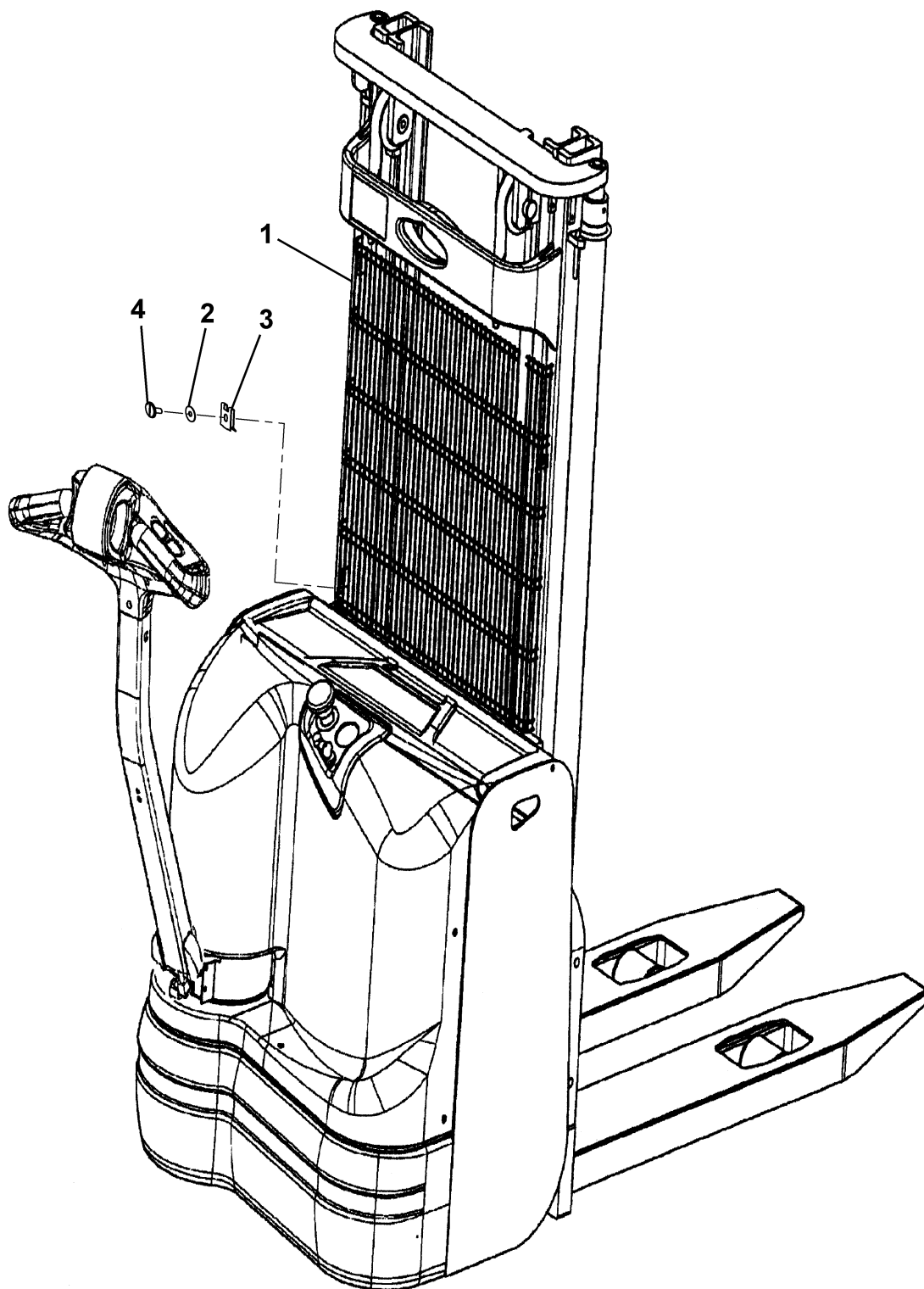
BP170542

ITEM	PART NO.	DESCRIPTION	QTY
1	2039180	Cylinder Assy (Lift Height=4625mm)	1
1	2042267	Cylinder Assy (Lift Height=5395mm)	2
2	893268	Flow Valve	2
3	2041531	Kit-Seal	2
4	2039141	Cylinder Assy (Lift Height=4625mm)	1
4	2039143	Cylinder Assy (Lift Height=5395mm)	1
5	893278	Flow Valve	1
6	2041532	Kit-Seal	1

NOTES

[illegible]

WIRE NET (FIGURE 6-3) THREE STAGE MAST



BP290133

**WIRE NET
THREE STAGE MAST**

ITEM	PART NO.	DESCRIPTION	QTY
1	1477510	Guard (Lift Height=4625mm, 4795mm)	1
1	1482416	Guard (Lift Height=5395mm)	1
2	1608780	Washer	2
3	1585574	Nut , Clip	2
4	1587878	Screw	2

[illegible]

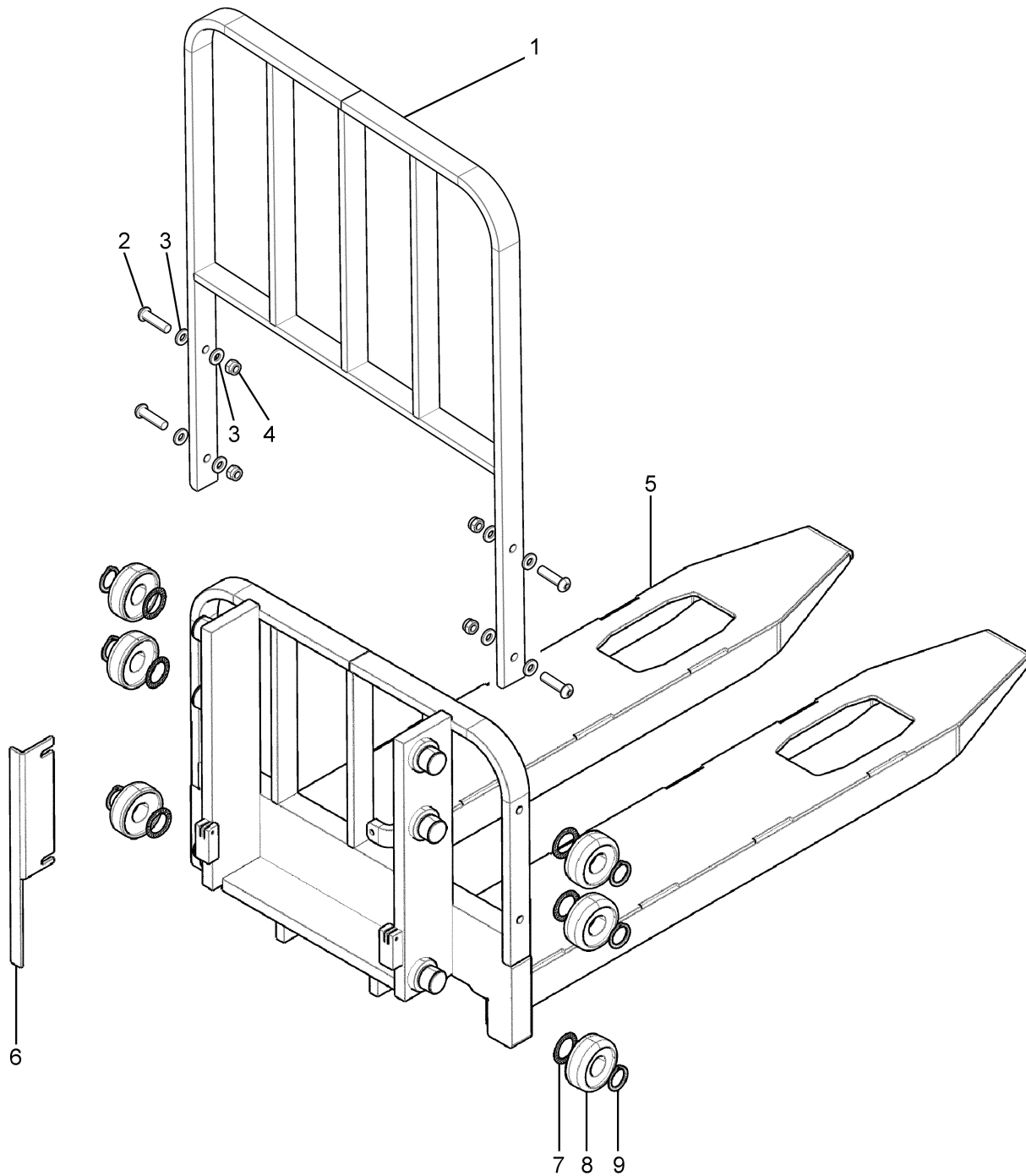
7

CARRIAGE

**SECTION 7
CARRIAGE**

DESCRIPTION	PAGE
FORKS CARRIAGE	
THREE STAGE FULL FREE-LIFT MAST	7-4
TWO STAGE LIMITED FREE-LIFT MAST	7-2

FORKS CARRIAGE (FIGURE 7-1) TWO STAGE LIMITED FREE-LIFT MAST



BP120898

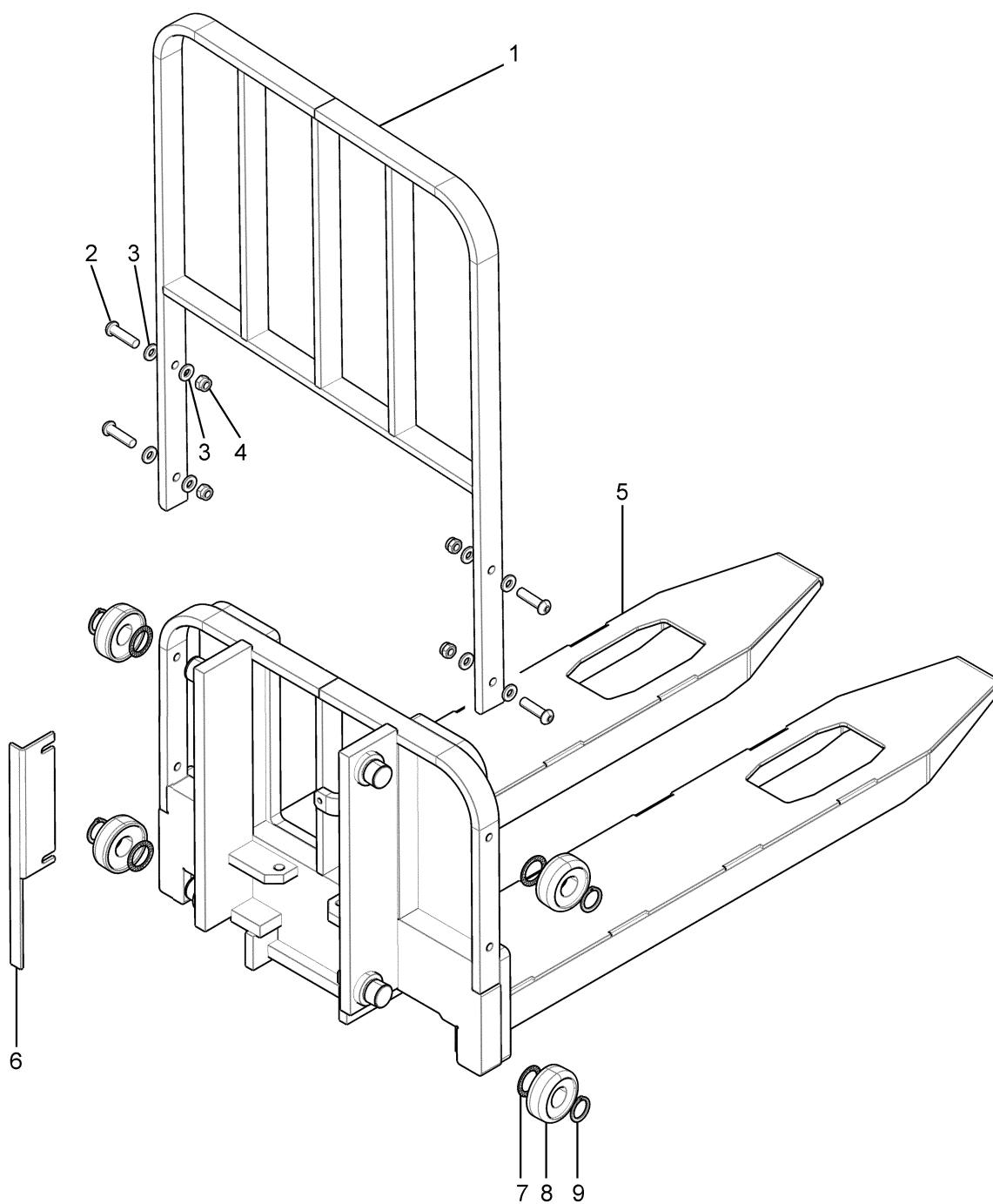
FORKS CARRIAGE **TWO STAGE LIMITED FREE-LIFT MAST**

ITEM	PART NO.	DESCRIPTION	QTY
1	1609012	Load Backrest	1
2	296184	Screw	4
3	320161	Washer	8
4	316837	Nut	4
5	1464121	Forks Carriage	1
7	129387	Shim (0.020 in.)	AR
8	1363272	Load Roller ("B") (3.24 in.)	6
9	58901	Snap Ring	6
10	a 388049		
	292682	Lockwasher	1
11	a 2795475	Screw	1

FOOTNOTES:

a Not Illustrated

FORKS CARRIAGE (FIGURE 7-2) THREE STAGE FULL FREE-LIFT MAST



BP120899

**FORKS CARRIAGE
THREE STAGE FULL FREE-LIFT MAST**

ITEM	PART NO.	DESCRIPTION	QTY
1	1609012	Load Backrest	1
2	296184	Screw	4
3	320161	Washer	8
4	316837	Nut	4
5	1464122	Forks Carriage	1
7	129387	Shim (0.020 in.)	AR
8	1363272	Load Roller ("B") (3.24 in.)	4
9	58901	Snap Ring	4
10	a 388049		
	292682	Lockwasher	1
11	a 2795475	Screw	1

FOOTNOTES:**a** Not Illustrated

[illegible]

**SECTION 8
NUMERICAL INDEX**

PART NO.	PAGE	ITEM	PART NO.	PAGE	ITEM	PART NO.	PAGE	ITEM	PART NO.	PAGE	ITEM
12928	1-11	12	292685	5-3	20	320161	1-9	28	1470148	1-9	10
12929	1-11	11		6-3	40		7-3	3	1470169	4-5	15
12981	1-9	5	293656	1-7	2		7-5	3	1470170	4-5	6
14560	4-3	5	293727	1-15	15	320169	5-3	10	1470171	4-5	11
15201	5-3	4	293742	2-3	18		6-3	19	1470172	4-5	
	6-3	4	293761	1-17	12	339530	2-3	14	1470185	1-11	3
15262	5-3	22	293774	1-3	4	355859	4-5	8	1471485	2-7	30
	6-3	11		5-3	24	355866	4-5	16	1471487	2-7	31
16447	4-3	7		6-3	13	370646	2-3	24	1476664	1-3	12
17218	4-3	10	293796	5-3	25	370647	2-3	25	1476666	1-3	9
17319	4-3	9		6-3	23	388048	1-11	17	1477510	6-7	1
58901	5-3	16	296011	1-9	25	886889	1-12	2	1480077	1-7	10
	6-3	36	296017	1-15	16	893268	5-4	3	1482416	6-7	1
	7-3	9	296027	1-15	3		6-4	2	1482417	5-7	1
	7-5	9	296075	5-3	21	893278	6-4	5	1482587	4-3	11
58908	6-3	31		6-3	39	1150086	4-5	3	1482608	4-3	8
67551	1-9	6	296128	1-15	2	1330410	2-7	29	1484420	6-3	32
104506	6-3	43	296153	2-3	22	1330420	2-7	21	1491574	1-9	27
128469	1-7	6	296184	7-3	2	1330438	2-7	22	1495150	4-5	5
129387	6-3	34		7-5	2	1337828	2-3	23	1497753	5-3	27
	7-3	7	296192	1-17	8	1343989	6-3	14		6-3	25
	7-5	7	296215	5-3	14	1363272	5-3	15	1498102	1-17	28
131357	5-3	9	296240	1-9	21		6-3	35	1499660	1-17	1
	6-3	18	296418	1-9	1		7-3	8	1503997	4-5	14
131716	6-3	41	296509	2-3	10		7-5	8	1503998	4-5	9
136687	1-11	16	302398	2-3	28	1369409	2-7	20	1508206	2-7	13
136749	1-5	1	302399	1-5	8	1464121	7-3	5	1508211	2-7	11
194723	3-2	12		2-3	29	1464122	7-5	5	1508232	2-7	15
215212	6-3	42	302400	1-5	5	1465597	5-3	1	1520400	2-7	27
222016	1-3	23	302404	5-3	7		6-3	1	1524569	1-12	4
239503	1-9	2		6-3	7	1465598	6-3	10	1525904	1-15	8
	2-3	17	302452	3-2	3	1466588	1-3	17	1530189	3-2	14
287851	6-3	30	302456	3-2	1	1466595	1-17	4	1530390	1-7	3
287854	4-5	17	304522	5-3	29	1466606	1-17	21	1531075	1-17	7
287860	1-9	35		6-3	37	1466607	1-17	23	1531785	3-2	9
292500	2-3	34	310433	5-3	28	1466935	1-17	17	1531787	3-2	10
292501	2-3	35		6-3	38	1467779	1-17	5	1531788	3-2	11
292518	1-9	34	316837	7-3	4	1468053	6-3	9	1532113	1-9	27
292528	1-9	29		7-5	4	1468055	6-3	9	1532284	2-4	4
292533	6-3	17	320152	1-17	2	1468058	6-3	8	1532285	2-4	1
292680	5-3	23	320154	1-15	15	1468060	6-3	8	1532286	2-4	6
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	6-3	24	320157	1-5	2	1470097	1-9	7	1539161	1-9	27
292681	6-3	29		1-15	16	1470099	1-9	8	1539448	1-9	17
292682	1-9	33		2-3	16	1470100	1-9	9	1539449	1-9	3
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	7-5	10		1-15	1	1470106	1-9	24	1547629	2-7	12
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NUMERICAL INDEX (continued)

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1581353	1-17	25		6-4	3	2770500	1-9	27		6-3	6
1585574	5-7	3	2041532	6-4	6	2770501	1-9	27	2794101	5-3	17
	6-7	3	2042267	6-4	1	2770502	1-9	27		6-3	27
1587878	5-7	4	2043556	1-17	29	2770642	1-15	16	2794114	1-15	14
	6-7	4	2045629	5-3	11	2770656	3-2	6	2794592	1-3	8
1588252	2-3	21	2045976	1-7	8	2770675	4-5	13	2794669	1-9	18
1589277	2-3	27	2048593	2-7	18	2770677	4-5		2794733	1-9	12
1590162	2-3	26	2048835	2-7	28	2770679	4-5	19	2795455	6-3	22
1591525	1-11	18	2056243	2-7	24	2770681	4-3	8	2795456	5-3	13
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1600793	4-5	10	2059401	2-4	7	2770690	4-3	6		7-5	11
1606701	4-5	12	2059402	2-4	8	2770775	1-3	18	2795502	1-5	6
1608758	2-7	9	2059812	2-7	16	2770785	4-5	1	2795583	2-7	6
1608759	2-7	3	2060217	1-9		2770786	4-5	2	2798661	4-3	2
1608780	5-7	2	2060555	2-7	1	2770825	1-7	9	2798663	4-3	1
	6-7	2	2060614	2-7	26	2780616	1-3	16	3041219	3-2	13
1609012	7-3	1	2067917	2-7	14	2787339	2-7	4	4002239	2-7	33
	7-5	1	2067918	2-7	8	2787340	2-7	10	4004119	4-5	20
1610944	2-7	23	2069534	1-12	8	2787341	2-7	5	4011993	1-17	26
1619955	1-3	19	2097314	1-17	14	2787345	2-7	7	4011997	1-17	22
1626085	1-11	4	2106374	1-11		2787350	2-7	2	4012949	1-11	9
1648884	2-3	19	2108279	2-4	13	2790487	2-3	8	4013193	1-17	18
1652960	1-17	15	2108280	2-4	14	2790663	5-3	2	4013194	1-17	3
1656304	2-7	32	2108281	2-4	15		6-3	2	4013196	1-17	13
1660875	1-17	6	2108282	2-4	10	2790866	1-15	17	4020725	2-7	34
1682988	1-12	6	2108283	2-4	12	2791931	1-15	11	4020726	2-7	35
1682989	1-12	7	2108284	2-4	11		3-2	8	4021991	4-5	4
1692973	2-3	33	2108354	2-3	38	2791933	1-9	15	4023318	1-3	24
1693742	2-3	7	2108492	2-4	9	2791942	3-2	5	4033244	1-3	3
1694086	1-7	1	2108634	2-3	20	2791951	6-3	20	4034229	6-3	15
1694816	1-7	7	2111838	1-17	11	2791975	1-9	22	4036431	1-17	16
1694817	1-7	5	2302743	1-3	6	2791987	1-15	12	4036439	1-17	27
1694818	1-7	4	2302852	2-7	17	2792009	1-9	14	4036580	1-17	10
1698418	1-3	13	2309579	2-7	19	2792053	1-15	10	4038296	2-3	12
1700498	1-11	14	2770149	6-3	33	2792105	1-9	16	4041497	2-3	13
1700499	1-11	1	2770157	5-3	12	2792165	1-9	26	4046682	1-17	24
1700500	1-11	13		6-3	21	2792219	3-2	7	4055221	1-5	7
1700501	1-11	2	2770276	5-3	18	2792265	3-2	4	4075554	1-12	5
1700503	1-11	7		6-3	26	2792295	1-15	9	4075556	1-12	3
1700504	1-11	8	2770277	5-3	19	2792314	1-9	19	4084426	1-17	
1700505	1-11	6		6-3	44	2792378	1-15	7	4096945	2-3	15
1700506	1-11	15	2770353	1-15	6	2792743	1-3	5	4151227	1-5	12
1700507	1-11	5	2770354	1-7	10	2792746	1-3	2	4151231	1-5	11
1701031	1-11	10	2770370	3-2	2	2793258	1-9	13	4151233	1-5	13
1706800	1-17	9	2770378	1-5	9	2793383	5-3	3	4151377	1-15	4
2035276	2-7	25		1-15	5		6-3	3	4151566	1-3	7
2039141	6-4	4		2-3	31	2793761	1-3	1	4151710	6-3	28
2039143	6-4	4	2770380	1-3	22	2793783	1-3	21	4152691	1-5	3
2039180	6-4	1	2770386	1-3	10	2793850	5-3	5	4601335	1-15	13
2039183	5-4	1	2770396	4-3	3		6-3	5	4611400	1-17	20

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